Group 04

MEDI-CINE

Version 1.0

Revision History

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Use case Specification Document

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Use Case Diagram

1. Browse Movies and View Movie Details

1.1 Browse Movies and View Movie Details

1.1.1 Brief Description

Primary Actor: Customer
The Customer browses movies currently showing or coming soon, searches or filters the movie catalogue, and views details of a selected movie, including available showtimes.

1.2 Flow of Events

1.2.1 Basic Flow

1. The Customer opens the movie catalogue page.
2. The system displays movies marked as now-showing or coming-soon, with available catalogue information such as title, poster, genre, duration, and age rating.
3. The Customer optionally enters a search term or selects available filters.
4. The system applies the search/filter criteria and displays matching movies.
5. The Customer selects a movie.
6. The system displays the selected movie's details, poster/trailer when available, and available showtimes.
7. The Customer selects a showtime to continue to seat selection.

1.2.2 Alternative Flows

1.2.2.1 No Movies Available in the Catalogue

Trigger: The system retrieves the catalogue, but no movies are available to display.

1. At Step 4 of the Basic Flow, the system finds no movies matching the specified search term or filters.
2. The system displays a message indicating that no matching movies were found.
3. The Customer may modify the search term or filters.
4. If the Customer modifies the criteria, the flow resumes at Step 4 of the Basic Flow. Otherwise, the use case ends.

1.2.2.2 No Available Showtimes

Trigger: At Step 2 of the Basic Flow, the system retrieves the catalogue, but no movies are available to display.

1. At Step 6 of the Basic Flow, the system determines that no showtimes are available for the selected movie.
2. The system displays the movie details and indicates that no showtimes are currently available.
3. The Customer may return to the movie catalogue and select another movie.
4. The use case ends, or the flow resumes when the Customer selects another movie.

1.2.2.3 Movie Details Cannot Be Retrieved

Trigger: At Step 4 of the Basic Flow, the system finds no movies matching the specified search term or filters.

1. At Step 6 of the Basic Flow, the system fails to retrieve the selected movie's details or available showtimes.
2. The system displays an error message indicating that the information cannot currently be retrieved.
3. The Customer may retry the request or return to the movie catalogue.
4. If the Customer retries, the flow resumes at Step 6 of the Basic Flow. Otherwise, the use case ends.

1.3 Special Requirements

1.3.1 Clear Presentation and Usability

Movie catalogue, showing status (now-showing / coming-soon), movie details (title, genre, duration, poster, trailer, age rating), and showtime information must be presented clearly with readable contrast and labels.

1.3.2 Responsive Design and Browser Compatibility

The customer-facing web application must be responsive across desktop, tablet, and mobile browsers, and support current versions of Chrome, Edge, and Firefox.

1.3.3 Data Authority and Security

The backend system is the authoritative source for movie catalogue and showtime data. All communication between the client interface and backend REST APIs must occur over HTTPS.

1.3.4 Response Time (Performance)

At least 95% of movie catalogue and showtime retrieval requests must complete within 2 seconds under expected workload conditions.

1.4 Preconditions

1.4.1 System and Catalogue Availability

The MEDI-CINE web application and backend REST APIs are operational and accessible over HTTPS.

1.4.2 Configured Operational Data

Movie records and active showtimes have been configured and maintained in the system by the Administrator (note: the Customer does not need to be logged in to browse movies or view details).

1.5 Postconditions

1.5.1 Success End Condition

The Customer has viewed the desired movie details and available showtimes, or has selected an active showtime to proceed to the seat selection flow

1.5.2 Data Integrity (No State Modification)

No movie records, showtime schedules, seat states (AVAILABLE, HELD, SOLD), or booking records are modified during this use case.

1.6 Extension Points

1.6.1 Seat Selection and Temporary Hold

Location: Basic Flow, Step 7 (when the Customer selects an available showtime to continue to seat selection).

1.6.2 View and Submit Movie Reviews / Ratings

Location: Basic Flow, Step 6 (while viewing the selected movie's details, eligible Customers may view or submit movie reviews and ratings)

1.6.3 AI Movie / Showtime Assistant (Optional Extension)

Location: Basic Flow, Steps 1–4 (the Customer may use natural-language queries to discover movies and showtimes grounded in approved MEDI-CINE data).

2. Select Seats

2.1 Select Seats

2.1.1 Brief Description

Primary Actor: Customer
The Customer views the real-time seat map for a selected showtime, chooses one or more available seats, and requests to proceed with booking. The system performs a server-side availability re-check, places a time-limited temporary hold (HELD) on the selected seats to prevent concurrent double-booking, and calculates the provisional ticket price before transitioning to the booking and payment flow.

2.2 Flow of Events

2.2.1 Basic Flow

1. This use case begins when the Customer selects an available showtime from the movie details page.
2. The system retrieves and displays the screening room information, the real-time seat map, seat types (e.g., regular, VIP, Sweetbox), pricing rules, and current seat states (AVAILABLE, HELD, SOLD).
3. The Customer selects one or more AVAILABLE seats on the seat map.
4. The system highlights the selected seats and updates the booking summary, displaying the selected seat identifiers, seat types, and total calculated price based on seat type and showtime category.
5. The Customer confirms the seat selection to proceed to booking.
6. The system verifies that the Customer is authenticated and performs an authoritative server-side check to confirm that all selected seats are still AVAILABLE.
7. The system atomically changes the state of the selected seats from AVAILABLE to HELD with a defined expiration timer, propagates the updated seat state to other active clients, and proceeds to the Book Movie Tickets use case.

2.2.2 Alternative Flows

2.2.2.1 Customer Is Not Logged In

Trigger: At Step 5 or Step 6 of the Basic Flow, an unauthenticated Customer attempts to confirm seat selection and hold seats.

1. The system prompts the Customer to log in or register an account (invoking Authentication and Authorization).
2. If the Customer successfully authenticates, the flow resumes at Step 6 of the Basic Flow to re-check seat availability.
3. If the Customer cancels authentication, the system remains on the seat map without holding the seats, and the use case ends.

2.2.2.2 Selected Seat Becomes Unavailable (Concurrent Conflict)

Trigger: At Step 6 of the Basic Flow, the server-side availability check detects that one or more selected seats have already been changed to HELD or SOLD by another concurrent Customer.

1. The system rejects the hold request for the entire selection and does not lock any of the requested seats.
2. The system refreshes the seat map to reflect the latest authoritative seat states (HELD or SOLD).
3. The system displays a clear notification informing the Customer that the selected seat(s) are no longer available.
4. The Customer may select different AVAILABLE seats, resuming the flow at Step 3 of the Basic Flow, or exit the use case.

2.2.2.3 No Seats Available for the Selected Showtime (Sold Out)

Trigger: At Step 2 of the Basic Flow, all seats in the screening room for the selected showtime are currently HELD, SOLD, or unavailable.

1. The system displays the seat map with all seats marked as unavailable and indicates that no seats are currently available for selection.

2. The Customer may return to the movie details page to select a different showtime.
3. The use case ends.

2.2.2.4 Customer Deselects or Changes Seats Before Confirmation

Trigger: Between Steps 3 and 5 of the Basic Flow, the Customer clicks on a previously selected seat or chooses a different seat before confirming the hold.

1. The system removes the deselected seat from the selection and recalculates the total price in the booking summary.
2. The flow continues at Step 3 of the Basic Flow.

2.2.2.5 Temporary Seat Hold Expires

Trigger: After Step 7 of the Basic Flow, the temporary hold timer expires before the booking and payment process is completed.

1. The system automatically releases the held seats, reverting their state from HELD back to AVAILABLE, and propagates the updated state to the seat map.
2. The system notifies the Customer that the seat hold has expired.
3. The flow may resume at Step 2 of the Basic Flow if the Customer wishes to select seats again; otherwise, the use case ends.

2.3 Special Requirements

2.3.1 Concurrency and Consistency

Seat availability checks and state transitions (AVAILABLE → HELD) must be handled transactionally on the backend server. The system must handle at least 50 concurrent booking attempts without confirming or holding the same seat for more than one Customer.

2.3.2 Usability and Accessibility

The seat map must provide understandable seat-state indicators, clear price summaries, readable contrast, labels, and non-color-only seat cues so Customers can easily distinguish between AVAILABLE, HELD, SOLD, and selected seats across desktop and mobile screen sizes.

2.3.3 Fault Tolerance on Expired Holds

Timed-out or abandoned holds must automatically release seats back to AVAILABLE without leaving seats permanently locked.

2.3.4 Live Seat-State Propagation

Changes in seat states (AVAILABLE, HELD, SOLD) must be propagated to other viewing Customers with low delay.

2.4 Preconditions

2.4.1 Valid Showtime Selected

The Customer has selected an active, scheduled showtime associated with a configured screening room, seat layout, and ticket pricing policy

2.4.2 Authoritative Backend Service

The backend service, database, and seat-hold/cache mechanism are operational over HTTPS.

2.5 Postconditions

2.5.1 Success End Condition

The selected seats are locked in the HELD state on the backend for a limited duration associated with the Customer's session, the updated seat states are visible to other users, and the Customer is ready to create a booking (Use-Case 3: Book Movie Tickets).

2.5.2 Failure / Cancellation End Condition

If seat selection fails due to concurrent conflict, unauthenticated cancellation, or hold expiration, no seats remain locked in the HELD state for this Customer's request, and authoritative seat availability remains consistent.

2.6 Extension Points

2.6.1 Book Movie Tickets

Location: Basic Flow, Step 7 (immediately after selected seats are validated and placed in the HELD state, the flow continues to booking creation and discount/promotion handling).

3. Book Movie Tickets

3.1 Book Movie Tickets

3.1.1 Brief Description

Primary Actor: Customer
The Customer reviews the booking summary for the temporarily held seats, optionally applies a promotion or discount code, and confirms the creation of a movie ticket booking. The system validates the active seat hold and pricing on the backend, creates a booking record in the PENDING state, and manages the booking lifecycle across PENDING, SUCCESS, FAILED, and EXPIRED states in coordination with the payment and seat-hold mechanisms.

3.2 Flow of Events

3.2.1 Basic Flow

1. This use case begins after the system successfully places a temporary hold (HELD) on the Customer's selected seats.
2. The system displays the booking review screen, showing the selected movie title, screening room, showtime, held seat identifiers and seat types, itemized ticket prices, hold countdown timer, and an option to enter a discount code.
3. The Customer optionally enters a promotion or discount code and requests validation.
4. The system validates the promotional rules or discount code on the backend and updates the total payable amount in the booking summary.
5. The Customer confirms the booking details and chooses to proceed to online payment.
6. The system verifies server-side that the Customer's temporary seat hold (HELD) has not expired and that the seat and price data remain valid.
7. The system creates an auditable booking record with an initial status of PENDING, associates it with the Customer's account and the HELD seats, and transitions to the Make Payment use case.
8. Upon receiving confirmation of a successful payment outcome from the Make Payment use case, the system updates the booking status from PENDING to SUCCESS and transitions the associated seats from HELD to SOLD.
9. The system records the completed transaction in the Customer's booking history, sends a booking confirmation notification to the Customer, and triggers the Retrieve E-Ticket / Validate E-Ticket flow.

3.2.2 Alternative Flows

3.2.2.1 Invalid or Inapplicable Discount Code

Trigger: At Step 4 of the Basic Flow, the Customer submits a discount code that is invalid, expired, or does not meet promotional rules.

1. The system rejects the discount code and displays a clear validation message explaining that the code cannot be applied.
2. The system retains the original authoritative ticket price without applying a discount.
3. The Customer may enter a different discount code or proceed without a discount, resuming the flow at Step 3 or Step 5 of the Basic Flow.

3.2.2.2 Temporary Seat Hold Expires Before Booking Confirmation

Trigger: Between Steps 2 and 6 of the Basic Flow, the temporary seat-hold timer expires before the Customer confirms the booking.

1. The system prevents the creation of a new active booking and marks any draft/initiated booking attempt as EXPIRED.
2. The system automatically releases the held seats from HELD back to AVAILABLE and propagates the updated seat states to the seat map.
3. The system notifies the Customer that the seat hold has expired and redirects the Customer back to the seat

selection page.
4. The use case ends.

3.2.2.3 Customer Cancels Booking Before Payment

Trigger: At any point prior to Step 7 of the Basic Flow, the Customer chooses to cancel the booking or return to the seat map to change seats.

1. The system cancels the current booking request without creating a confirmed booking.
2. The system immediately releases the Customer's HELD seats back to AVAILABLE and updates the live seat map.
3. The Customer is returned to the Select Seats use case, and this use case ends.

3.2.2.4 Booking Fails or Expires During Payment Processing

Trigger: At Step 8 of the Basic Flow, the Make Payment use case returns a failed payment outcome or a payment timeout/expiration.

1. The system updates the booking status from PENDING to FAILED (if payment was declined/cancelled) or EXPIRED (if the payment window timed out).
2. The system atomically releases the associated seats from HELD back to AVAILABLE so they are not permanently locked.
3. The system ensures that no QR e-ticket is generated for the failed or expired booking.
4. The system records the FAILED or EXPIRED status in the Customer's booking history, notifies the Customer of the booking outcome, and the use case ends.

3.2.2.5 Duplicate Booking Request Submission

Trigger: At Step 5 of the Basic Flow, the Customer submits duplicate booking creation requests (e.g., due to double-clicking or network retries) for the same held seats.

1. The system detects that a PENDING booking already exists for the Customer's active seat hold and rejects or deduplicates the redundant request.
2. The system continues with the single existing PENDING booking at Step 7 of the Basic Flow.

3.3 Special Requirements

3.3.1 Authoritative Pricing and State Consistency

The backend must act as the single authoritative source for ticket prices, promotional discount calculations, seat availability, and booking state transitions (PENDING, SUCCESS, FAILED, EXPIRED).

3.3.2 Transactional Integrity and No Unpaid Tickets

Booking state updates and seat state transitions (HELD → SOLD or HELD →AVAILABLE) must execute transactionally. Under no circumstances may a FAILED or EXPIRED booking leave seats permanently locked or issue an unpaid QR ticket.

3.3.3 Clear Status Feedback and Usability

The user interface must display clear price summaries, discount validation messages, hold expiration countdowns, and explicit booking status feedback across desktop and mobile browsers.

3.3.4 Security and Access Control

Booking creation and booking history access require an authenticated Customer session over HTTPS with server-side input validation.

3.4 Preconditions

3.4.1 Authenticated Customer

The Customer is logged into the MEDI-CINE system with a valid Customer role.

3.4.2 Active Temporary Seat Hold

The Customer has one or more seats locked in the HELD state for a valid, active showtime, and the hold expiration timer has not elapsed.

3.4.3 Configured Pricing Rules

Ticket pricing by seat type and showtime category, along with any active promotional rules, are configured in the backend database.

3.5 Postconditions

3.5.1 Success End Condition

A booking record is stored in the database and transitions from PENDING to SUCCESS upon payment confirmation; the associated seats are marked SOLD; the booking is visible in the Customer's Booking History; and a booking confirmation notification is triggered.

3.5.2 Failure / Expiration End Condition

If the hold expires or payment fails/times out, the booking record is marked as FAILED or EXPIRED, all associated HELD seats are released back to AVAILABLE, and no e-ticket is issued.

3.6 Extension Points

3.6.1 Apply Promotions and Discount Codes

Location: Basic Flow, Steps 3–4 (when the Customer enters a promotional or discount code to adjust the booking total).

3.6.2 Make Payment

Location: Basic Flow, Step 7 (once the booking record is created in the PENDING state, the system invokes payment processing).

3.6.3 Issue E-Ticket and Send Notifications

Location: Basic Flow, Step 9 (once the booking reaches the SUCCESS state, the system issues the QR e-ticket and sends confirmation/reminder notifications).

4. Make Payment

4.1 Make Payment

4.1.1 Brief Description

Primary Actor: Customer
Secondary Actor: External Payment Provider
The Customer initiates and completes an online payment for a pending booking through an external payment provider (sandbox gateway). The system generates a payment request, processes the payment outcome reported by the provider, updates the payment and booking states (PENDING, SUCCESS, FAILED, or EXPIRED), updates the corresponding seat states (SOLD or AVAILABLE), and triggers QR e-ticket issuance only upon confirmed payment.

4.2 Flow of Events

4.2.1 Basic Flow

1. This use case begins when the Customer confirms a PENDING booking with active HELD seats and chooses to proceed to online payment.
2. The system verifies on the backend that the booking is still in the PENDING state, the temporary seat hold has not expired, and the total payable amount is authoritative.
3. The system creates a payment transaction record in the PENDING state and invokes the payment adapter to generate a secure payment session with the External Payment Provider.
4. The system redirects the Customer to the External Payment Provider interface (or displays the sandbox payment prompt) with the transaction details.
5. The Customer submits valid payment credentials and authorizes the transaction on the External Payment Provider interface.
6. The External Payment Provider processes the transaction and sends a payment confirmation callback/response to the MEDI-CINE backend.
7. The system validates the authenticity of the callback, verifies that the transaction has not already been processed, and updates both the payment status and booking status from PENDING to SUCCESS.
8. The system atomically updates the state of the associated seats from HELD to SOLD and propagates the seat-state update.
9. The system records the confirmed payment in the Customer's booking history, sends a payment confirmation notification, and displays the payment success screen to proceed to Retrieve E-Ticket / Validate E-Ticket.

4.2.2 Alternative Flows

4.2.2.1 Payment Declined or Cancelled by Customer (Payment Failed)

Trigger: At Step 5 or Step 6 of the Basic Flow, the Customer cancels the payment on the gateway, or the External Payment Provider declines the transaction and reports a failure outcome.

1. The External Payment Provider sends a failed payment response/callback to the system.
2. The system validates the response and updates the payment status and booking status from PENDING to FAILED.
3. The system atomically releases the associated seats from HELD back to AVAILABLE and updates the live seat map so the seats are not permanently locked.
4. The system ensures that no QR e-ticket is issued for the booking.
5. The system displays a clear payment failure notification to the Customer, records the FAILED status in the booking history, and the use case ends.

4.2.2.2 Payment Timeout or Temporary Seat Hold Expires (Payment Expired)

Trigger: Between Steps 4 and 6 of the Basic Flow, the Customer does not complete the payment before the payment timeout or the temporary seat-hold timer expires.

1. The system detects the expiration (via backend timer or timeout callback) and updates the payment status and booking status from PENDING to EXPIRED.
2. The system automatically releases the associated seats from HELD back to AVAILABLE and propagates the updated state to the seat map.
3. The system ensures that no unpaid QR e-ticket is generated.
4. The system informs the Customer that the payment session has expired and prompts the Customer to restart seat selection if they still wish to buy tickets.
5. The use case ends.

4.2.2.3 Duplicate Payment Callback or Duplicate Request

Trigger: At Step 3 or Step 6 of the Basic Flow, the system receives duplicate payment initiation requests from the Customer or duplicate payment confirmation callbacks from the External Payment Provider for the same booking.

1. The system inspects the authoritative transaction state on the backend and identifies that the payment request or callback has already been processed.
2. The system ignores the redundant request/callback without creating duplicate payment records or issuing duplicate QR e-tickets.
3. The system returns the existing authoritative transaction status and continues normal execution.

4.2.2.4 External Payment Provider Unreachable or Callback Error

Trigger: At Step 3 or Step 6 of the Basic Flow, the system cannot establish a connection with the External Payment Provider, or the payment callback fails/is corrupted.

1. The system logs the integration error without exposing payment secrets or sensitive user data.
2. If the failure occurs before payment authorization (Step 3), the system notifies the Customer that the payment service is temporarily unavailable and allows the Customer to retry while the seat hold remains active.
3. If the hold expires before a valid confirmation is received, the system transitions the booking to EXPIRED and releases the HELD seats back to AVAILABLE so seats are never permanently locked.
4. The use case ends.

4.3 Special Requirements

4.3.1 Transactional Consistency and Idempotency

Payment state updates (PENDING, SUCCESS, FAILED, EXPIRED), booking states, and seat states (HELD, SOLD, AVAILABLE) must remain strictly consistent under concurrent access and duplicate callbacks. Duplicate payment callbacks must never generate duplicate tickets.

4.3.2 Reliability and Fault Tolerance

The system must handle payment failures, timeouts, expired holds, and failed callbacks safely without leaving seats permanently locked in the HELD state or issuing unpaid e-tickets.

4.3.3 Security and Secret Protection

All payment requests and callbacks must be transmitted over HTTPS and validated on the backend. Payment gateway secrets, credentials, and sensitive user data must be protected and must never be exposed to the client browser or written to system logs.

4.3.4 Modularity (Payment Adapter Isolation)

Provider-specific payment logic must be isolated behind a backend payment adapter so changes to the external payment gateway (e.g., VNPay, MoMo, or card sandbox) do not disrupt core booking logic.

4.3.5 Clear Payment Status Feedback

The customer interface must provide clear, unambiguous feedback for pending, successful, failed, and expired payment states across desktop and mobile browsers.

4.4 Preconditions

4.4.1 Active Pending Booking and Held Seats

The Customer is authenticated, has an active booking in the PENDING state, and the associated seats are currently locked in the HELD state within the allowed hold duration.

4.4.2 Payment Sandbox / Provider Availability

The backend payment module and the External Payment Provider (sandbox environment) are configured and accessible over HTTPS.

4.5 Postconditions

4.5.1 Success End Condition

The payment and booking states are recorded as SUCCESS; the held seats are permanently marked as SOLD for that showtime; the transaction is logged in the Customer's booking history; a payment update notification is sent; and a valid QR e-ticket is ready for issuance.

4.5.2 Failure or Expiration End Condition

The payment and booking states are recorded as FAILED or EXPIRED; all associated seats are released from HELD back to AVAILABLE; and no valid QR e-ticket is issued.

4.6 Extension Points

4.6.1 Retrieve E-Ticket / Validate E-Ticket

Location: Basic Flow, Step 9 (immediately after payment is confirmed as SUCCESS, the system generates the QR e-ticket and makes it available to the Customer).

5. Retrieve E-Ticket / Validate E-Ticket

5.1 Retrieve E-Ticket / Validate E-Ticket

5.1.1 Brief Description

Primary Actors: Customer, Cinema Staff
Following a confirmed payment, the system issues a unique QR-coded e-ticket associated with the confirmed booking. The Customer retrieves and views the e-ticket via the customer web interface or booking history and presents the QR code at cinema admission. Cinema Staff use the staff dashboard to scan or enter the QR ticket identifier, verify that the ticket is paid and unused, and record admission by marking the ticket status as USED to enforce one-time entry.

5.2 Flow of Events

5.2.1 Basic Flow

1. This use case begins when a booking payment is confirmed as SUCCESS, or when an authenticated Customer accesses a confirmed booking from the booking history page.
2. The system generates a unique e-ticket containing a QR code, ticket identifier, movie title, screening room, showtime, confirmed seat numbers, payment status, and an initial ticket state of unused/valid, and sends an e-ticket notification to the Customer.
3. The Customer opens the e-ticket details on a mobile or desktop browser and presents the QR code (or ticket identifier) to the Cinema Staff at the cinema admission point.
4. The Cinema Staff accesses the admission validation screen on the desktop dashboard and scans the QR code or enters the QR ticket identifier.
5. The system queries the authoritative backend to verify the ticket identifier, confirming that the associated booking payment is SUCCESS, the showtime is valid, and the ticket has not yet been marked as USED.
6. The system displays a clear valid/paid admission result to the Cinema Staff along with the movie, showtime, room, and seat details.
7. The Cinema Staff confirms admission for the Customer.
8. The system atomically updates the e-ticket status to USED, records the admission timestamp, and displays a confirmation that check-in is complete.

5.2.2 Alternative Flows

5.2.2.1 E-Ticket Has Already Been Used (Duplicate Admission Attempt)

Trigger: At Step 5 of the Basic Flow, the backend check determines that the scanned or entered QR e-ticket is already marked as USED.

1. The system immediately rejects the admission validation request.
2. The system displays a prominent warning message on the staff dashboard indicating that the ticket has already been used, along with the previous admission record.
3. The Cinema Staff denies repeated entry for that ticket, and the use case ends.

5.2.2.2 Invalid, Forged, or Non-Existent QR Code

Trigger: At Step 5 of the Basic Flow, the scanned QR code or entered ticket identifier does not match any confirmed e-ticket record in the backend database.

1. The system rejects the lookup request.
2. The system displays an "Invalid Ticket" message on the staff dashboard.
3. The Cinema Staff may re-scan the QR code or manually verify the ticket identifier with the Customer, resuming at Step 4 of the Basic Flow; otherwise, admission is denied and the use case ends.

5.2.2.3 Booking Is Unpaid, Failed, or Expired

Trigger: At Step 1 of the Basic Flow (when a Customer attempts to retrieve an e-ticket from booking history) or at Step 5 of the Basic Flow (during staff lookup), the associated booking is in the PENDING, FAILED, or EXPIRED state.

1. For the Customer: the system displays the current booking/payment status in the booking history without displaying a valid QR e-ticket.
2. For the Cinema Staff: if a reference code is entered for an unpaid or cancelled booking, the system displays a clear notice that no paid/valid ticket exists for that booking and blocks check-in.
3. The use case ends.

5.2.2.4 Wrong Showtime or Cancelled Showtime at Admission

Trigger: At Step 5 of the Basic Flow, the ticket is paid and unused, but the associated showtime does not match the current screening schedule or has been cancelled.

1. The system alerts the Cinema Staff that the ticket belongs to a different showtime (or a cancelled showtime) and displays the scheduled date and time.
2. The system does not mark the ticket as USED so the ticket remains valid for its actual scheduled showtime (if upcoming).
3. The use case ends.

5.2.2.5 QR Camera Scan Fails (Manual Identifier Entry)

Trigger: At Step 4 of the Basic Flow, the QR code cannot be scanned due to screen brightness, camera issues, or a damaged display.

1. The Cinema Staff switches to manual entry on the staff dashboard and types the alphanumeric QR ticket identifier shown on the Customer's e-ticket.
2. The flow resumes at Step 5 of the Basic Flow.

5.3 Special Requirements

5.3.1 Strict One-Time Admission and State Consistency

A QR e-ticket must be generated only after confirmed payment (SUCCESS) and can be validated for admission only once. Once marked USED, the ticket state is permanent and must be rejected on any subsequent validation attempt, including concurrent scan requests.

5.3.2 Fast Validation Performance

QR ticket lookup and admission validation at check-in must complete without noticeable delay, with a target response time of within 2 seconds so cinema entry queues are not slowed down.

5.3.3 Role-Based Access Control and Security

E-ticket retrieval is restricted to the authenticated Customer who owns the booking, while QR validation and marking a ticket as USED are restricted strictly to authenticated Cinema Staff (or Administrator) roles over HTTPS.

5.3.4 Cross-Device Usability

The Customer's e-ticket and QR code must render clearly on both mobile phone and desktop browsers. The staff validation interface must be optimized for desktop browsers (Chrome, Edge, Firefox) with clear visual indicators distinguishing valid/unused tickets from used or invalid tickets.

5.4 Preconditions

5.4.1 For Retrieving an E-Ticket (Customer)

The Customer is authenticated, and the booking has completed payment confirmation with a SUCCESS status and seats marked as SOLD.

5.4.2 For Validating an E-Ticket (Cinema Staff)

The Cinema Staff member is logged into the staff dashboard with authorized role permissions, and the backend database service is accessible over HTTPS.

5.5 Postconditions

5.5.1 E-Ticket Retrieval Success End Condition

The Customer has viewed or downloaded the confirmed e-ticket and QR code from the booking confirmation screen or booking history without altering the ticket's unused status.

5.5.2 Admission Validation Success End Condition

The valid e-ticket is permanently marked as USED in the authoritative backend database, and the admission check-in record is saved.

5.5.3 Validation Rejection End Condition

If the ticket is invalid, unpaid, or already marked USED, admission is rejected and the authoritative state in the database remains unchanged.

5.6 Extension Points

5.6.1 Submit Movie Review and Rating

Location: After Step 8 of the Basic Flow (once a Customer has purchased a ticket and attended the screening, the eligible Customer may submit a movie review or rating).

5.6.2 Upcoming Showtime Reminders

Location: After Step 2 of the Basic Flow (prior to the scheduled showtime, the system may send automated email or notification reminders for active, unused e-tickets).

6. Register, Log In, and Manage Profile

6.1 Register, Log In, and Manage Profile

6.1.1 Brief Description

Primary Actors: Customer, Staff, Admin
Unregistered users can register for a new Customer account by providing only the information required for account creation and ticket booking. Registered users (Customer, Cinema Staff, and Administrator) can log in to establish an authenticated session with role-based access control and log out to terminate their session. Authenticated Customers can also view and update their personal profile information.

6.2 Flow of Events

6.2.1 Basic Flow

1. This use case begins when an unauthenticated user selects the option to register or log in, or when an authenticated user accesses their account profile or chooses to log out.
2. **Account Registration (for new Customers):**

- a. The user selects the registration option on the customer web interface.
- b. The system displays the registration form requesting only the essential personal information needed for accounts and bookings.
- c. The user enters the required details and submits the registration request.
- d. The system validates the input fields on the backend, verifies that the account identifier is unique, securely stores the protected credentials, and creates a new account assigned with the Customer role.

3. **User Log In and Role-Based Routing:**

- a. The user opens the login page, enters their account credentials, and submits the login request.
- b. The system validates the inputs and verifies the credentials against the authoritative backend database over HTTPS.
- c. The system creates a protected authenticated session, determines the user's assigned role (Customer, Cinema Staff, or Administrator), and routes the user to the appropriate interface (the responsive customer web interface for Customers, the admission validation dashboard for Cinema Staff, or the administration dashboard for Administrators).

4. **Profile Management (for authenticated Customers):**

- a. The authenticated Customer navigates to the profile management page.
- b. The system retrieves and displays the Customer's current profile details.
- c. The Customer edits their profile information (or updates their credentials) and submits the changes.
- d. The system validates the updated inputs on the backend, saves the changes to the database, and displays a clear confirmation message.

5. **User Log Out:**

- a. The authenticated user selects the logout option.
- b. The system invalidates the active session data, clears authentication state, and redirects the user to the public home or login page.

6.2.2 Alternative Flows

6.2.2.1 Invalid Input Format or Missing Required Fields

Trigger: At Step 1d, Step 2b, or Step 3d of the Basic Flow, the submitted registration, login, or profile update data fails server-side input validation.

1. The system rejects the request without modifying the database.
2. The system displays clear validation messages indicating which fields are missing or invalid.
3. The user corrects the inputs and resubmits the form, resuming the flow at the corresponding step; otherwise, the use case ends.

6.2.2.2 Account Already Exists During Registration

Trigger: At Step 1d of the Basic Flow, the backend detects that the submitted email or account identifier is already registered in the system.

1. The system rejects the registration attempt and displays a validation message informing the user that the account already exists.
2. The user may enter a different identifier to continue registration at Step 1c or switch to the login page at Step 2a.

6.2.2.3 Invalid Login Credentials or Deactivated Account

Trigger: At Step 2b of the Basic Flow, the submitted credentials do not match any active account in the backend database.

1. The system denies authentication without exposing sensitive credential details.
2. The system displays a clear error message stating that the login credentials are invalid (or the account is inactive).
3. The user may re-enter their credentials to resume at Step 2a, or exit the use case.

6.2.2.4 Unauthorized Role Access Attempt

Trigger: After Step 2c of the Basic Flow, an authenticated user (or unauthenticated visitor) attempts to access a protected page or invoke a backend API restricted to another role (e.g., a Customer or Cinema Staff attempting to access Administrator operations, or a Customer attempting to access Staff QR validation).

1. The backend role-based access control check intercepts and denies the unauthorized request.
2. The system displays an access-denied message (or redirects an unauthenticated user to the login page) and leaves operational data unchanged.
3. The use case ends.

6.3 Special Requirements

6.3.1 Security, Role-Based Access Control, and Secret Protection

The system must enforce role-based access control on all backend REST APIs over HTTPS, validate all user inputs server-side, and protect user credentials and session data without ever logging secrets. Unauthorized roles must be strictly denied access to protected administrative and staff operations in access-control tests.

6.3.2 Data Privacy and Minimal Collection

The system must collect only personal information strictly needed for user accounts and ticket bookings, and must avoid sending unnecessary personal data to any external service.

6.3.3 Compatibility and Responsive Usability

Customer registration, login, and profile pages must be responsive across mobile, tablet, and desktop browsers with readable contrast, clear labels, and understandable validation messages. Staff and Administrator login and dashboard access must support current versions of Chrome, Edge, and Firefox on desktop.

6.4 Preconditions

6.4.1 Operational Backend and Database

The MEDI-CINE web interface, authentication backend module, and database are operational and accessible over HTTPS.

6.4.2 Pre-Configured Staff and Administrator Accounts

For Cinema Staff and Administrator login flows, the respective user accounts and role permissions (Staff or Admin) have been provisioned in the system by an Administrator.

6.5 Postconditions

6.5.1 Registration Success End Condition

A new Customer account record with protected credentials and the Customer role is persisted in the database.

6.5.2 Login Success End Condition

An authenticated, role-bound session is established for the user (Customer, Cinema Staff, or Administrator), granting access only to the capabilities permitted for that role.

6.5.3 Profile Update Success End Condition

The Customer's updated profile information is validated and saved in the authoritative backend database.

6.5.4 Logout or Failure End Condition

Upon logout, session data is invalidated and protected operations are no longer accessible; upon failed registration, failed login, or unauthorized access attempts, no unauthorized session is created and existing account data remains unchanged.

6.6 Extension Points

6.6.1 Resume Seat Selection and Booking Flow

Location: Basic Flow, Step 2c (when an unauthenticated Customer logs in or registers after being prompted during Select Seats, the system redirects the Customer back to complete their seat hold and booking).

6.6.2 View Booking History

Location: Basic Flow, Step 3b (from the authenticated Customer's account/profile area, the Customer can navigate to view current and past bookings, payment statuses, and QR e-tickets).

7. Manage Movie Catalogue

7.1 Manage Movie Catalogue

7.1.1 Brief Description

Primary Actor: Administrator
The Administrator uses the desktop administration dashboard to create, view, edit, and deactivate (or remove) movie records in the cinema catalogue. The system enforces role-based access control, validates movie metadata (including title, genre, duration, poster, trailer, age rating, and showing status), and persists the updated catalogue in the authoritative backend database so it can be used for showtime scheduling and customer movie discovery.

7.2 Flow of Events

7.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Movie Catalogue Management section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions and retrieves the current list of movie records from the backend database, displaying each movie's title, genre, duration, age rating, poster/trailer status, and showing status (now-showing, coming-soon, or deactivated).
3. **Create a New Movie Record:**

- a. The Administrator selects the option to add a new movie.
- b. The system displays the movie creation form.
- c. The Administrator enters the movie details, including title, genre, duration, poster, trailer, age rating, and initial showing status (now-showing or coming-soon), and submits the form.
- d. The system validates the submitted inputs on the backend, creates a new movie record in the database, logs the state change, and displays a confirmation message along with the updated movie catalogue.

4. **Edit an Existing Movie Record:**

- a. The Administrator selects an existing movie from the catalogue and chooses the edit option.
- b. The system retrieves and displays the selected movie's current details in an editable form.
- c. The Administrator modifies the desired fields (such as title, genre, duration, poster, trailer, age rating, or showing status) and submits the changes.
- d. The system validates the updated data on the backend, updates the movie record in the database, and displays a confirmation message.

5. **Deactivate (or Remove) a Movie Record:**

- a. The Administrator selects a movie record and chooses to deactivate (or remove) it from active showing.
- b. The system checks whether the movie can be safely deactivated or removed without corrupting active showtimes or existing customer bookings, updates the movie's showing status to deactivated/inactive (or removes an unreferenced draft record), and confirms the update to the Administrator.

7.2.2 Alternative Flows

7.2.2.1 Invalid or Missing Movie Information

Trigger: At Step 3d or Step 4d of the Basic Flow, the server-side validation detects missing required fields or invalid values (e.g., an empty title, a non-positive duration, an unsupported age rating, or a malformed poster/trailer URL/file).

1. The system rejects the submission without saving changes to the database.
2. The system displays clear validation messages highlighting the specific fields that require correction.
3. The Administrator corrects the input values and resubmits the form to resume at Step 3d or Step 4d, or cancels the operation.

7.2.2.2 Duplicate Movie Record Detection

Trigger: At Step 3d of the Basic Flow, the backend detects that a movie with the same title and identifying attributes

already exists in the catalogue.

1. The system prevents the creation of a duplicate record and displays a validation message notifying the Administrator of the existing movie entry.
2. The Administrator may edit the existing movie record instead (resuming at Step 4) or modify the input details.

7.2.2.3 Movie Has Active Showtimes or Confirmed Bookings Upon Deactivation/Removal

Trigger: At Step 5b of the Basic Flow, the Administrator attempts to remove or deactivate a movie that is currently linked to active upcoming showtimes or confirmed customer bookings/tickets.

1. The system prevents hard deletion of the movie record to preserve operational and historical booking data consistency.
2. If active upcoming showtimes exist, the system displays a warning message informing the Administrator that the associated showtimes must be cancelled or completed first (or marks the movie as deactivated for new scheduling while preserving historical booking references).
3. The use case continues at Step 2 of the Basic Flow.

7.2.2.4 Duration Modification Conflicts with Scheduled Showtimes

Trigger: At Step 4d of the Basic Flow, the Administrator increases a movie's duration while the movie is already assigned to active showtimes, and the new duration would cause a room-time overlap with subsequent showtimes.

1. The system rejects the duration change and displays a validation message explaining the schedule overlap conflict.
2. The Administrator may adjust the showtime schedule first or cancel the duration edit, and the flow returns to Step 4b.

7.2.2.5 Unauthorized Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct API call), a user without the Administrator role attempts to view or modify the administrative movie catalogue endpoints.

1. The backend role-based access control mechanism denies the request.
2. The system displays an access-denied message and logs the unauthorized attempt without altering any movie data.
3. The use case ends.

7.3 Special Requirements

7.3.1 Role-Based Access Control and Input Validation

All movie creation, modification, and deactivation operations must be restricted strictly to authenticated Administrator roles on backend REST APIs over HTTPS, with server-side validation for all inputs.

7.3.2 Data Consistency and Operational Integrity

The backend database is the single authoritative source for movie metadata and showing statuses. Deactivating or editing a movie must never invalidate or corrupt historical booking records, issued QR e-tickets, or revenue reports.

7.3.3 Desktop Dashboard Compatibility and Usability

The administration movie management interface must be optimized for desktop browsers supporting current versions of Chrome, Edge, and Firefox, and must provide clear validation messages and status indicators.

7.3.4 Observability and Auditability

Meaningful state changes (such as creating a movie, updating showing status, or deactivating a movie) and validation errors must be recorded by the backend without logging sensitive secrets.

7.4 Preconditions

7.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE administration dashboard with an active, role-verified Administrator session.

7.4.2 Operational Backend and Persistence

The backend REST APIs and database environment are operational over HTTPS.

7.5 Postconditions

7.5.1 Success End Condition

The newly created, updated, or deactivated movie record is persisted in the authoritative backend database; active movies (now-showing and coming-soon) are immediately reflected in customer movie discovery (Browse Movies and View Movie Details) and available for showtime scheduling.

7.5.2 Failure / Rejection End Condition

If validation fails, a schedule conflict is detected, or access is unauthorized, no invalid changes are committed to the database and the movie catalogue remains in its previous consistent state.

7.6 Extension Points

7.6.1 Schedule Showtimes

Location: After Step 3d or Step 4d of the Basic Flow (once an active movie record with a valid duration is saved in the catalogue, the Administrator can proceed to create or update showtimes for that movie).

8. Manage Cinema Rooms and Seats

8.1 Manage Cinema Rooms and Seats

8.1.1 Brief Description

Primary Actor: Administrator
The Administrator uses the desktop administration dashboard to configure and maintain screening rooms (supporting a mid-size cinema operation of approximately 15–20 screening rooms), room formats (2D, 3D, IMAX), seating layouts, seat types (regular, VIP, Sweetbox), and operational seat statuses. The system enforces role-based access control, validates layout consistency against active showtimes and bookings, and stores the authoritative room and seat configurations in the backend database.

8.2 Flow of Events

8.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Cinema Room and Seat Management section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions and retrieves the list of configured screening rooms, displaying each room's name/identifier, format (2D, 3D, IMAX), total seat capacity, and operational status.
3. **Create and Configure a Screening Room:**

- a. The Administrator selects the option to create a new screening room.
- b. The system displays the room configuration form.
- c. The Administrator enters the room identifier/name, selects the screening format (2D, 3D, or IMAX), defines the seating grid dimensions (rows and columns), and submits the room configuration.
- d. The system validates the room details on the backend, creates the screening room record, and generates an initial seat layout grid.

4. **Configure Seat Layout, Seat Types, and Seat Status:**

- a. The Administrator selects a screening room to configure its seating layout.
- b. The system displays the interactive seat map editor showing all seats in the room, their row/number labels, current seat types (regular, VIP, Sweetbox), and operational seat statuses.
- c. The Administrator assigns or modifies seat types (regular, VIP, or Sweetbox) and operational statuses (e.g., active/available for scheduling or inactive/out-of-service) for individual seats or seat rows, and submits the layout changes.
- d. The system validates the seat layout on the backend, verifies that no active showtime bookings are violated, saves the updated seat configuration in the database, and displays a confirmation message.

5. **Update or Deactivate a Screening Room:**

- a. The Administrator selects an existing screening room to update its format/details or deactivate it from scheduling.
- b. The system verifies that the room has no conflicting active showtimes, persists the updated room configuration in the database, logs the change, and confirms the update to the Administrator.

8.2.2 Alternative Flows

8.2.2.1 Invalid Room Details or Invalid Seat Layout Configuration

Trigger: At Step 3d or Step 4d of the Basic Flow, server-side validation detects missing or invalid inputs (e.g., missing room name, unsupported room format, zero/negative row or column counts, or invalid seat type assignment).

1. The system rejects the configuration request without modifying the database.
2. The system displays clear validation messages indicating the specific fields or seat entries that must be corrected.
3. The Administrator corrects the input data and resubmits, resuming the flow at Step 3c or Step 4c; otherwise, the use case ends.

8.2.2.2 Duplicate Screening Room Identifier

Trigger: At Step 3d or Step 5b of the Basic Flow, the backend detects that another screening room with the same name or identifier already exists in the cinema.

1. The system prevents saving the duplicate room name and displays a clear validation message.
2. The Administrator enters a unique room identifier and resubmits the form.

8.2.2.3 Seat or Room Modification Conflicts with Active Showtimes or Booked Seats

Trigger: At Step 4d or Step 5b of the Basic Flow, the Administrator attempts to remove seats, change structural seat layouts, or deactivate a room/seat that currently has active showtimes with seats in the HELD or SOLD state.

1. The system blocks the destructive layout modification or room deactivation to protect existing customer bookings and seat-state consistency.
2. The system displays a validation warning identifying the affected upcoming showtimes or booked seats.
3. The Administrator must wait until the active showtimes are completed (or cancel/reschedule unbooked showtimes first) before applying structural layout changes, and the flow returns to Step 4b or Step 2.

8.2.2.4 Unauthorized Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct API request), a user without the Administrator role attempts to view or modify screening room and seat configurations.

1. The backend role-based access control mechanism denies the request.
2. The system returns an access-denied message and logs the unauthorized attempt without changing any room or seat records.
3. The use case ends.

8.3 Special Requirements

8.3.1 Data Consistency and Protection of Active Bookings

The backend database is the authoritative source for room formats, seat layouts, seat types, and seat statuses. Structural modifications to rooms or seats must never corrupt active showtimes, invalidate HELD or SOLD seats, or alter historical booking and occupancy report records.

8.3.2 Operational Scale Support

The room and seat management module must smoothly support configuration and maintenance for a mid-size cinema operation of approximately 15–20 screening rooms with diverse formats (2D, 3D, IMAX) and seat types (regular, VIP, Sweetbox).

8.3.3 Role-Based Access Control and Input Validation

All room and seat configuration APIs must be restricted strictly to authenticated Administrator roles over HTTPS and validate all layout parameters server-side.

8.3.4 Desktop Dashboard Usability and Accessibility

The interactive seat layout editor on the administration dashboard must support current desktop versions of Chrome, Edge, and Firefox, providing readable contrast, clear seat labels, non-color-only cues for seat types (regular, VIP, Sweetbox) and statuses, and clear validation messages.

8.4 Preconditions

8.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE desktop administration dashboard with an active, role-verified Administrator account.

8.4.2 Operational Backend and Database

The backend REST APIs and database persistence layer are operational over HTTPS.

8.5 Postconditions

8.5.1 Success End Condition

The created or updated screening room, format (2D, 3D, IMAX), seating layout, seat types (regular, VIP, Sweetbox), and operational seat statuses are saved in the authoritative backend database and are immediately available for showtime scheduling, ticket pricing rules, and real-time seat map rendering.

8.5.2 Failure / Conflict End Condition

If input validation fails, a conflict with active showtimes/bookings occurs, or access is unauthorized, no invalid changes are persisted and existing room and seat configurations remain unchanged.

8.6 Extension Points

8.6.1 Configure Ticket Pricing

Location: After Step 4d of the Basic Flow (once seat types such as regular, VIP, and Sweetbox are configured for the screening rooms, the Administrator can configure or update ticket pricing rules by seat type and day/time category).

8.6.2 Schedule Showtimes

Location: After Step 3d or Step 4d of the Basic Flow (once an active screening room with a valid seat layout is saved, the Administrator can schedule movie showtimes in that room).

9. Manage Showtime Schedule

9.1 Manage Showtime Schedule

9.1.1 Brief Description

Primary Actor: Administrator

The Administrator uses the desktop administration dashboard to create, update, activate, or cancel movie showtimes across configured screening rooms. The system enforces role-based access control, automatically calculates screening time windows using the selected movie's duration and required room-cleaning intervals, validates against room-time overlaps, links the showtime to applicable pricing categories, and initializes seat availability for active showtimes in the authoritative backend database.

9.2 Flow of Events

9.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Showtime Scheduling section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions and retrieves the current showtime schedule, displaying scheduled showtimes filtered by date, screening room, movie title, start/end time, format, and showtime status.
3. **Create a New Showtime:**

- a. The Administrator selects the option to create a new showtime.
- b. The system displays the showtime scheduling form containing selectable active movies, active screening rooms (with their formats: 2D, 3D, IMAX), date/time pickers, and day/time pricing categories (e.g., weekday, weekend, early showtime).
- c. The Administrator selects a movie, chooses a screening room, specifies the screening date and start time, confirms the pricing category, and submits the request.
- d. The system calculates the total occupied room window using the movie's duration plus the required cleaning interval, and validates server-side that no room-time overlap exists in the selected screening room.
- e. The system saves the new showtime record in the database, initializes the showtime's seat availability map based on the room's configured seat layout (setting active seats to AVAILABLE), and displays a confirmation message.

4. **Update or Activate an Existing Showtime:**

- a. The Administrator selects an existing showtime from the schedule to modify its timing/room details or activate it for customer booking.
- b. The system validates that the updated schedule does not overlap with other showtimes in the target room and does not conflict with existing customer seat holds or bookings.
- c. The system updates the showtime status or details in the database, logs the change, and confirms the update to the Administrator.

5. **Cancel a Showtime:**

- a. The Administrator selects a scheduled showtime and chooses the option to cancel it.
- b. The system updates the showtime status to cancelled, removes the showtime from customer movie discovery and new booking flows, and releases the room time slot for future scheduling.

9.2.2 Alternative Flows

9.2.2.1 Room-Time Overlap or Insufficient Cleaning Interval Detected

Trigger: At Step 3d or Step 4b of the Basic Flow, the backend validation determines that the proposed showtime window (movie duration plus cleaning interval) overlaps with another scheduled showtime in the same screening room.

1. The system rejects the showtime creation or update request to prevent schedule conflicts.
2. The system displays a clear validation message detailing the conflicting showtime, screening room, and

occupied time range.

3. The Administrator adjusts the start time or selects a different screening room and resubmits the form, resuming at Step 3c or Step 4a; otherwise, the use case ends.

9.2.2.2 Invalid Schedule Parameters or Inactive Movie/Room Selected

Trigger: At Step 3d or Step 4b of the Basic Flow, the Administrator specifies a start time in the past, leaves required fields blank, selects an incompatible room format, or selects a movie/room that has been deactivated.

1. The system rejects the request without modifying the database.
2. The system displays clear validation messages indicating the invalid fields or inactive references.
3. The Administrator corrects the inputs and resubmits, resuming at Step 3c or Step 4a.

9.2.2.3 Rescheduling or Cancelling a Showtime That Has Held or Sold Seats

Trigger: At Step 4b or Step 5a of the Basic Flow, the Administrator attempts to change the room/time or cancel an active showtime that already has seats in the HELD or SOLD state.

1. If the Administrator attempts to change the screening room or time, the system blocks the modification to prevent inconsistent booking states and seat assignment conflicts.
2. If the Administrator chooses to cancel the showtime, the system displays a confirmation warning showing the number of affected bookings/tickets; upon confirmation, the system marks the showtime as cancelled, blocks any further seat selection or admission check-in for that showtime, and triggers customer notifications (and administrative refund/status handling).
3. The flow returns to Step 2 of the Basic Flow.

9.2.2.4 Unauthorized Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct API call), a user without the Administrator role attempts to create, edit, activate, or cancel showtimes.

1. The backend role-based access control mechanism denies the request.
2. The system displays an access-denied message and logs the unauthorized attempt without modifying the schedule.
3. The use case ends.

9.3 Special Requirements

9.3.1 Authoritative Overlap and Cleaning Interval Validation

All schedule conflict checks—including movie duration and mandatory cleaning intervals between screenings in the same room—must be enforced server-side on the authoritative backend database. No two active showtimes in the same screening room may have overlapping time windows.

9.3.2 Task Efficiency and Desktop Usability

The showtime scheduling interface on the desktop administration dashboard must support current versions of Chrome, Edge, and Firefox, present clear validation messages and schedule timelines, and enable an Administrator to create a showtime within the target cycle time of 3 minutes.

9.3.3 Data Consistency with Seat Maps and Bookings

Activating a showtime must consistently initialize its seat map from the screening room's layout so customers see accurate AVAILABLE seats, and any status change to a showtime must remain consistent with seat holds, bookings, tickets, and reports.

9.3.4 Security and Auditability

Showtime management APIs must be restricted strictly to authenticated Administrator roles over HTTPS, and all schedule creations, updates, activations, and cancellations must be logged for operational auditability.

9.4 Preconditions

9.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE administration dashboard with a verified Administrator role.

9.4.2 Configured Movies, Rooms, Seats, and Pricing Categories

At least one active movie record (with a defined duration) and one active screening room (with a configured seat layout and format) exist in the backend database.

9.5 Postconditions

9.5.1 Success End Condition

The new or updated showtime is stored in the authoritative backend database without room-time overlap; activated showtimes have an initialized seat map (AVAILABLE) and are immediately visible to Customers during movie discovery (Browse Movies and View Movie Details) and to Cinema Staff on the dashboard

9.5.2 Failure / Conflict End Condition

If a room-time overlap, cleaning-interval violation, invalid input, or unauthorized request occurs, the system rejects the change and preserves the existing schedule state.

9.6 Extension Points

9.6.1 Configure Ticket Pricing

Location: Basic Flow, Step 3c (when assigning or adjusting ticket pricing rules by seat type and showtime day/time category such as weekday, weekend, or early showtime).

9.6.2 Customer Notifications and Reminders

Location: Alternative Flow 9.2.2.3 (if an active showtime with confirmed bookings is cancelled, or prior to an upcoming active showtime, the system triggers notifications/reminders to affected Customers).

10. Manage Ticket Pricing

10.1 Manage Ticket Pricing

10.1.1 Brief Description

Primary Actor: Administrator

The Administrator uses the desktop administration dashboard to configure, view, and update ticket pricing rules by seat type (regular, VIP, Sweetbox) and day/time category (weekday, weekend, early showtime). The system enforces role-based access control, validates pricing inputs server-side, and stores the authoritative pricing configuration in the backend database so that accurate, transparent prices are calculated during customer seat selection and ticket booking.

10.2 Flow of Events

10.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Ticket Pricing Management section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions and retrieves the current ticket pricing rules from the backend database, displaying configured prices organized by seat type (regular, VIP, Sweetbox) and day/time category (weekday, weekend, early showtime).
3. **Create a New Ticket Pricing Rule:**

- a. The Administrator selects the option to add a new pricing rule.
- b. The system displays the ticket pricing configuration form with selectable seat types and day/time categories.
- c. The Administrator selects the target seat type (regular, VIP, or Sweetbox), selects the applicable day/time category (weekday, weekend, or early showtime), enters the ticket price amount, and submits the form.
- d. The system validates the submitted pricing rule on the backend, ensures no duplicate conflicting rule exists for the same seat-type and day/time combination, saves the pricing record in the database, logs the change, and displays a confirmation message.

4. **Update an Existing Ticket Pricing Rule:**

- a. The Administrator selects an existing pricing rule from the pricing table and chooses the edit option.
- b. The system displays the current seat type, day/time category, and price value in an editable form.
- c. The Administrator enters the updated price value and submits the changes.
- d. The system validates the updated price on the backend, updates the authoritative pricing record in the database for subsequent seat selections and bookings, logs the state change, and displays a confirmation message to the Administrator.

10.2.2 Alternative Flows

10.2.2.1 Invalid or Missing Price Input

Trigger: At Step 3d or Step 4d of the Basic Flow, server-side input validation detects that required fields are missing or the entered price value is invalid (e.g., negative, zero, or non-numeric).

1. The system rejects the pricing request without modifying the database.
2. The system displays clear validation messages indicating the invalid price field or missing selection.
3. The Administrator corrects the input and resubmits the form, resuming at Step 3c or Step 4c; otherwise, the use case ends.

10.2.2.2 Duplicate Pricing Rule Combination

Trigger: At Step 3d of the Basic Flow, the backend detects that an active pricing rule already exists for the selected seat type (regular, VIP, or Sweetbox) and day/time category (weekday, weekend, or early showtime) combination.

1. The system prevents the creation of a duplicate pricing rule to maintain consistent operational data.
2. The system displays a clear validation message informing the Administrator that a rule for this combination

already exists.

3. The Administrator may choose to edit the existing pricing rule instead (resuming at Step 4) or select a different combination at Step 3c.

10.2.2.3 Price Update Occurs While Bookings or Seat Holds Are Active

Trigger: At Step 4d of the Basic Flow, the Administrator updates a ticket price while there are existing PENDING bookings, SUCCESS bookings, or issued QR e-tickets for active showtimes.

1. The system applies the new pricing rule to subsequent customer seat selections and new bookings.
2. The system preserves the already committed price amounts recorded on existing PENDING or SUCCESS bookings, issued tickets, and historical revenue reports so operational data and payment states remain consistent.
3. The system completes the update and displays a confirmation message to the Administrator.

10.2.2.4 Unauthorized Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct backend API call), a user without the Administrator role attempts to view or modify ticket pricing configurations.

1. The backend role-based access control mechanism denies the request.
2. The system displays an access-denied message and logs the unauthorized attempt without altering any pricing data.
3. The use case ends.

10.3 Special Requirements

10.3.1 Authoritative Pricing and Data Consistency

The backend system must remain the single authoritative source for all ticket prices. Modifying ticket pricing rules must never alter the financial records of confirmed bookings, issued QR tickets, or historical revenue reports.

10.3.2 Role-Based Access Control and Input Validation

All ticket pricing creation and modification endpoints must be restricted strictly to authenticated Administrator roles over HTTPS, and all price inputs must be validated server-side.

10.3.3 Desktop Dashboard Compatibility and Usability

The pricing management interface must be optimized for desktop browsers (Chrome, Edge, and Firefox) and present clear labels, transparent price tables, and understandable validation messages.

10.3.4 Observability and Auditability

All creations and updates to ticket pricing policies must be logged by the backend for operational auditability without logging sensitive secrets.

10.4 Preconditions

10.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE desktop administration dashboard with a verified Administrator role.

10.4.2 Configured Seat Types and Day/Time Categories

Seat types (regular, VIP, Sweetbox) and showtime day/time categories (weekday, weekend, early showtime) are defined in the system.

10.5 Postconditions

10.5.1 Success End Condition

The newly created or updated ticket pricing rule is stored in the authoritative backend database and is immediately used to display transparent prices and calculate totals when Customers select seats (Select

Seats) and create bookings (Book Movie Tickets).

10.5.2 Failure / Rejection End Condition

If input validation fails, a duplicate rule conflict occurs, or access is unauthorized, no invalid changes are saved and the existing pricing configuration remains unchanged.

10.6 Extension Points

10.6.1 Manage Promotions and Discount Codes

Location: After Step 3d or Step 4d of the Basic Flow (once base ticket prices by seat type and day/time category are configured, the Administrator may configure promotional rules and discount codes that apply on top of the base ticket price).

11. View Booking History

11.1 View Booking History

11.1.1 Brief Description

Primary Actor: Customer

An authenticated Customer accesses the booking history page to view their current and previous ticket bookings. The system retrieves the Customer's transaction records from the authoritative backend database and displays booking and payment statuses (PENDING, SUCCESS, FAILED, EXPIRED), movie and showtime details, seat assignments, payment amounts, and associated e-ticket details, allowing the Customer to retrieve confirmed QR e-tickets.

11.2 Flow of Events

11.2.1 Basic Flow

1. This use case begins when an authenticated Customer navigates to the Booking History section on the customer web interface.
2. The system verifies the Customer's authenticated session and retrieves the list of current and past bookings associated strictly with that Customer's account from the backend database.
3. The system displays the booking history list ordered chronologically, showing summary information for each booking including the booking identifier, transaction date, movie title, screening room, showtime date/time, seat identifiers, total amount, and current booking/payment status (PENDING, SUCCESS, FAILED, or EXPIRED).
4. The Customer selects a specific booking record from the list to view full transaction details.
5. The system retrieves and displays the detailed booking view, including itemized seat types and prices, applied discounts (if any), payment status feedback, and ticket admission status (unused/valid or USED).
6. If the selected booking has a confirmed payment status (SUCCESS), the Customer selects the option to view the e-ticket, and the system displays the QR-coded e-ticket (invoking **Retrieve E-Ticket / Validate E-Ticket**).

11.2.2 Alternative Flows

11.2.2.1 No Previous or Current Bookings Found

Trigger: At Step 2 of the Basic Flow, the backend database query returns no booking records associated with the Customer's account.

1. The system displays a clear message indicating that the Customer has no current or past bookings.
2. The system provides a navigation link for the Customer to browse now-showing movies (Browse Movies and View Movie Details).
3. The use case ends.

11.2.2.2 Viewing a Failed or Expired Booking

Trigger: At Step 4 of the Basic Flow, the Customer selects a booking whose status is FAILED or EXPIRED.

1. The system displays the historical transaction details, including the movie, showtime, previously requested seats, and the FAILED or EXPIRED status reason.
2. The system clearly indicates that no QR e-ticket was issued for this booking and that the seats were released.
3. The Customer may return to the booking history list at Step 3 or navigate to the movie catalogue to start a new booking; otherwise, the use case ends.

11.2.2.3 Viewing an Active Pending Booking

Trigger: At Step 4 of the Basic Flow, the Customer selects a booking that is still in the PENDING state with an active temporary seat hold (HELD).

1. The system displays the booking details along with the remaining hold countdown timer and indicates that

payment is still pending (no QR e-ticket is shown yet).
2. If the seat hold has not expired, the Customer may choose to resume payment processing (transitioning to Make Payment).
3. If the hold timer has already elapsed upon inspection, the system updates the status to EXPIRED, releases the held seats to AVAILABLE, and refreshes the status display.

11.2.2.4 Unauthenticated Access Attempt or Expired Session

Trigger: At Step 1 of the Basic Flow, an unauthenticated user or a Customer with an expired session attempts to access the Booking History page.

1. The system denies direct access and redirects the user to the login page (Register, Log In, and Manage Profile).
2. Once the Customer successfully logs in, the flow resumes at Step 2 of the Basic Flow; otherwise, the use case ends.

11.2.2.5 Booking History Cannot Be Retrieved (Backend Error)

Trigger: At Step 2 or Step 5 of the Basic Flow, the system is unable to retrieve the booking records due to a temporary network or backend error.

1. The system displays a clear error message informing the Customer that booking history cannot be loaded at this time and provides a retry option.
2. If the Customer retries successfully, the flow resumes at Step 2 or Step 5; otherwise, the use case ends.

11.3 Special Requirements

11.3.1 Privacy, Security, and Role-Based Access Control

Booking history and e-ticket retrieval APIs must enforce strict authentication and authorization over HTTPS so that a Customer can view only their own bookings and tickets, and can never access another Customer's transaction records.

11.3.2 Authoritative Data and State Consistency

The displayed booking statuses (PENDING, SUCCESS, FAILED, EXPIRED), payment outcomes, seat assignments, and ticket states (unused or USED) must reflect the authoritative backend database state without discrepancy. Unpaid, failed, or expired bookings must never display a valid QR code.

11.3.3 Responsive Usability Across Devices

The booking history interface must be responsive across mobile phones, tablets, and desktop browsers (supporting current Chrome, Edge, Firefox, and mobile browsers), providing clear payment/ticket status feedback and readable contrast even under variable mobile network quality.

11.4 Preconditions

11.4.1 Authenticated Customer Session

The Customer is logged into the MEDI-CINE web application with an active, verified Customer account.

11.4.2 Operational Backend Service

The backend REST APIs and database persistence layer are operational over HTTPS.

11.5 Postconditions

11.5.1 Success End Condition

The Customer has viewed their list of current and past bookings, inspected specific booking and payment statuses, and optionally accessed the QR e-ticket for a confirmed (SUCCESS) booking.

11.5.2 Data Integrity (Read-Only Operation)

Viewing booking history does not modify confirmed bookings, payment records, SOLD seat states, or the unused/USED admission status of any issued e-ticket.

11.6 Extension Points

11.6.1 Retrieve E-Ticket / Validate E-Ticket

Location: Basic Flow, Step 6 (when the Customer opens a confirmed SUCCESS booking from their history to view or present the QR e-ticket for cinema admission).

11.6.2 Submit Movie Reviews and Ratings

Location: Basic Flow, Step 5 (from a completed booking for a movie the Customer has attended, the eligible Customer may proceed to submit a movie review or rating).

12. Validate E-Ticket and Record Admission

12.1 Validate E-Ticket and Record Admission

12.1.1 Brief Description

Primary Actor: Cinema Staff

Cinema Staff at the ticket counter or entrance lobby use the desktop validation dashboard to select or view relevant showtimes, scan a Customer's QR code via webcam/camera (or manually enter the QR ticket identifier), and verify ticket validity. The system checks the authoritative backend database to confirm that the ticket belongs to a confirmed paid booking and is unused, displays clear valid, used, or invalid results, and records one-time admission by marking the ticket status as USED.

12.2 Flow of Events

12.2.1 Basic Flow

1. This use case begins when a Customer arrives at the cinema entrance lobby before a showtime and presents a QR e-ticket to the Cinema Staff.
2. The authenticated Cinema Staff accesses the admission validation screen on the desktop dashboard and optionally views or filters the relevant upcoming showtime.
3. The Cinema Staff scans the Customer's QR code using the desktop webcam/camera scanner (or enters the QR ticket identifier into the lookup field).
4. The system transmits the QR ticket identifier over HTTPS to the authoritative backend service for validation.
5. The backend verifies that the ticket identifier exists, the associated booking payment status is SUCCESS, the showtime is active, and the ticket status is currently unused.
6. The system displays a clear VALID (paid and unused) validation result on the staff dashboard, showing the movie title, screening room, showtime, and assigned seat numbers.
7. The Cinema Staff confirms admission check-in for the Customer.
8. The system atomically updates the ticket status to USED in the backend database, records the auditable admission state, and displays a check-in completion confirmation so the Cinema Staff can admit the Customer.

12.2.2 Alternative Flows

12.2.2.1 Webcam/Camera Scan Fails Due to Low Light or Screen Glare (Manual Lookup)

Trigger: At Step 3 of the Basic Flow, the webcam/camera scanner cannot read the QR code due to low light in the entrance lobby or a dim/damaged phone screen.

1. The Cinema Staff types the QR ticket identifier directly into the manual lookup input field on the desktop dashboard and submits the lookup.
2. The flow resumes at Step 4 of the Basic Flow.

12.2.2.2 Ticket Has Already Been Used (Duplicate Admission / Reuse Attempt)

Trigger: At Step 5 of the Basic Flow, the backend check reveals that the ticket status is already marked as USED.

1. The system immediately rejects the admission attempt.
2. The system displays a prominent, unambiguous USED status warning on the staff dashboard along with the prior check-in details.
3. The Cinema Staff denies entry for the reused ticket, and the use case ends.

12.2.2.3 Invalid or Non-Existent QR Ticket Identifier

Trigger: At Step 5 of the Basic Flow, the scanned QR code or entered identifier does not match any issued e-ticket in the authoritative backend database.

1. The system rejects the validation request.
2. The system displays a clear INVALID result message on the staff dashboard.

3. The Cinema Staff may verify the code and retry scanning/entering at Step 3 of the Basic Flow; otherwise, admission is denied and the use case ends.

12.2.2.4 Associated Booking Is Unpaid, Failed, or Expired

Trigger: At Step 5 of the Basic Flow, the identifier corresponds to a booking whose payment state is PENDING, FAILED, or EXPIRED rather than SUCCESS.

1. The system blocks admission and displays an INVALID / UNPAID status result indicating that no confirmed paid ticket exists for the booking.
2. The Cinema Staff denies admission and directs the Customer to purchase a valid ticket, and the use case ends.

12.2.2.5 Ticket Belongs to a Different or Cancelled Showtime

Trigger: At Step 5 of the Basic Flow, the ticket is paid and unused, but its scheduled showtime does not match the current screening window or the showtime has been cancelled.

1. The system displays a validation warning indicating the mismatch (showing the ticket's actual scheduled date/time or cancelled status).
2. The system prevents marking a future showtime's ticket as USED prematurely so the ticket remains valid for its scheduled screening.
3. The use case ends.

12.2.2.6 Unauthorized Role Access Attempt

Trigger: At Step 2 or Step 4 of the Basic Flow, an unauthenticated user or a user with a Customer role attempts to access the staff validation dashboard or invoke the ticket check-in API.

1. The backend role-based access control mechanism denies the request.
2. The system displays an access-denied message without modifying any ticket state.
3. The use case ends.

12.3 Special Requirements

12.3.1 Response Time and Queue Efficiency (Performance)

Each QR lookup and admission validation must complete within **2 seconds per ticket** so that a single Cinema Staff member can process check-ins smoothly during peak pre-showtime lobby queues.

12.3.2 Strict One-Time Admission and Concurrency Consistency

The transition of a ticket from unused to USED must be atomic and consistent under concurrent access. A valid ticket can be used only once; any subsequent attempt to validate a ticket marked USED must be rejected.

12.3.3 Role-Based Security and Restricted Staff Permissions

The Cinema Staff account must be authenticated over HTTPS and restricted strictly to showtime viewing and ticket validation functions (preventing unauthorized access to administrative configuration functions).

12.3.4 Usability and Lobby Accessibility

The validation interface must run on current desktop versions of Chrome, Edge, and Firefox, support both webcam/camera QR scanning and manual code entry, and display high-contrast, unmistakable valid, used, and invalid visual results suitable for a busy, potentially low-light entrance lobby.

12.4 Preconditions

12.4.1 Authenticated Cinema Staff Session

The Cinema Staff member is logged into the desktop dashboard with an active, role-verified Staff (or Administrator) account.

12.4.2 Operational Backend and Camera/Input Hardware

The desktop browser (Chrome, Edge, or Firefox) has access to the backend HTTPS REST APIs and, for

QR scanning, a connected webcam/camera.

12.4.3 Issued QR E-Ticket

For a valid check-in, the Customer holds a QR e-ticket issued after a confirmed (SUCCESS) payment.

12.5 Postconditions

12.5.1 Success End Condition

The validated e-ticket is permanently marked as USED in the authoritative backend database, the admission check-in is recorded for operational and occupancy reporting, and the Customer is granted admission.

12.5.2 Rejection End Condition

If the ticket is already USED, INVALID, unpaid, or presented for the wrong showtime, admission is denied and the authoritative ticket state in the database remains accurate and uncorrupted.

12.6 Extension Points

12.6.1 Dashboard and Occupancy Reporting

Location: Basic Flow, Step 8 (recorded admission and ticket statuses contribute to showtime performance and seat occupancy statistics on the administration dashboard).

13. View Dashboard and Reports

13.1 View Dashboard and Reports

13.1.1 Brief Description

Primary Actor: Administrator

The Administrator accesses the desktop administration dashboard to view centralized operational statistics and sales reports, including total revenue, tickets sold, movie/showtime performance, and seat occupancy across the cinema's screening rooms. The system enforces role-based access control, aggregates authoritative booking, payment, seat, and admission records from the backend database, and presents accurate operational metrics without requiring manual spreadsheet compilation.

13.2 Flow of Events

13.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Dashboard and Reports section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions over HTTPS and queries the authoritative backend database for aggregated operational statistics.
3. The system displays the main summary dashboard showing key performance indicators: total revenue from confirmed payments, total tickets sold, active showtime count, and overall seat occupancy rate.
4. The Administrator optionally selects report filters or grouping criteria, such as a specific date range, movie title, screening room, or showtime.
5. The system validates the filter criteria on the backend, recalculates the metrics for the selected scope, and displays detailed breakdowns of:
   - Revenue and tickets sold by movie and by showtime.
   - Showtime performance and seat occupancy percentages (comparing SOLD seats and USED admission records against total room seat capacity).
6. The Administrator reviews the operational and financial performance data, and the use case ends.

13.2.2 Alternative Flows

13.2.2.1 No Operational or Sales Data Available for Selected Period

Trigger: At Step 3 or Step 5 of the Basic Flow, the backend query finds no confirmed bookings, sold tickets, or scheduled showtimes matching the selected date range, movie, or room filter.

1. The system displays zeroed summary counters ($0 revenue, 0 tickets sold, 0% occupancy) and a clear message indicating that no sales or showtime records exist for the selected criteria.
2. The Administrator may adjust or reset the filter criteria to resume at Step 4 of the Basic Flow; otherwise, the use case ends.

13.2.2.2 Invalid Filter Criteria

Trigger: At Step 4 of the Basic Flow, the Administrator submits an invalid filter combination (such as a start date later than the end date).

1. The system rejects the filter request and displays a clear validation message explaining the invalid date range or filter parameter.
2. The system retains the previously displayed report view and allows the Administrator to correct the filter inputs at Step 4.

13.2.2.3 Excluding Unpaid, Failed, or Expired Bookings from Revenue Metrics

Trigger: During data aggregation at Step 2 or Step 5 of the Basic Flow, the database contains bookings in PENDING, FAILED, or EXPIRED states, or seats temporarily in the HELD state.

1. The backend reporting logic strictly excludes PENDING, FAILED, and EXPIRED bookings as well as temporary HELD seats from confirmed revenue and sold-ticket totals.

2. Only bookings with confirmed SUCCESS payment states and SOLD seats are counted toward revenue and ticket sales, ensuring accurate financial reporting.
3. The flow continues normally at Step 3 or Step 5 of the Basic Flow.

13.2.2.4 Unauthorized Role Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct backend API request), an unauthenticated user, a Customer, or a Cinema Staff user attempts to access the administrative dashboard and financial reports.

1. The backend role-based access control mechanism intercepts and denies the request.
2. The system displays an access-denied message and logs the unauthorized attempt without exposing any revenue or operational data.
3. The use case ends.

13.3 Special Requirements

13.3.1 Data Accuracy and Authoritative Consistency

Revenue, tickets sold, showtime performance, and seat occupancy figures must be derived directly from the authoritative backend database and remain strictly consistent with confirmed payments (SUCCESS), seat states (SOLD), room capacities (across approximately 15–20 screening rooms), and check-in records (USED).

13.3.2 Role-Based Security and Confidentiality

Access to sales dashboards and operational reports must be restricted exclusively to authenticated Administrator roles over HTTPS. Unauthorized roles (Customer and Cinema Staff) must be denied access in access-control tests.

13.3.3 Desktop Browser Compatibility and Usability

The dashboard and reporting interface must be optimized for desktop browsers supporting current versions of Chrome, Edge, and Firefox, presenting readable contrast, clear labels, and understandable data summaries.

13.4 Preconditions

13.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE desktop administration dashboard with an active, role-verified Administrator session.

13.4.2 Operational Backend and Database

The backend reporting APIs and database persistence layer are operational over HTTPS.

13.5 Postconditions

13.5.1 Success End Condition

The Administrator has viewed accurate, up-to-date revenue, ticket sales, showtime performance, and seat occupancy reports matching the selected filter criteria.

13.5.2 Data Integrity (Read-Only Operation)

Viewing or filtering dashboard reports does not alter any movie, room, showtime, seat, booking, payment, or ticket records in the database.

13.6 Extension Points

13.6.1 Manage Showtime Schedule and Ticket Pricing

Location: After Step 5 of the Basic Flow (based on showtime performance and seat occupancy insights, the Administrator may navigate to adjust future showtime schedules or ticket pricing policies).

14. Submit Movie Reviews and Ratings

14.1 Submit Movie Reviews and Ratings

14.1.1 Brief Description

Primary Actor: Customer

An authenticated, eligible Customer submits a movie rating and written review feedback through the customer-facing web application. The system verifies the Customer's eligibility on the backend, validates the rating and review inputs, stores the feedback in the authoritative backend database, and updates the movie's displayed reviews and ratings.

14.2 Flow of Events

14.2.1 Basic Flow

1. This use case begins when an authenticated Customer views a movie's details page (or selects a completed booking from their booking history) and chooses the option to submit a movie review and rating.
2. The system verifies the Customer's authenticated session and checks the authoritative backend database to confirm that the Customer is eligible to review the selected movie.
3. The system displays the review and rating submission form for the selected movie.
4. The Customer selects a rating score, optionally enters written review feedback, and submits the form.
5. The system validates the submitted rating and review content on the backend over HTTPS.
6. The system saves the review and rating record in the backend database associated with the Customer's account and the selected movie, logs the state change, and updates the movie's aggregated rating display.
7. The system displays a clear confirmation message to the Customer along with the published feedback on the movie details view.

14.2.2 Alternative Flows

14.2.2.1 Customer Is Not Eligible to Submit a Review

Trigger: At Step 2 of the Basic Flow, the backend eligibility check determines that the Customer does not meet the eligibility criteria for the selected movie (e.g., the Customer has no confirmed SUCCESS booking or attended ticket for the movie).

1. The system prevents access to the review submission action.
2. The system displays a clear message explaining that only eligible Customers who have booked the movie can submit a review or rating.
3. The Customer remains on the read-only movie details view, and the use case ends.

14.2.2.2 Invalid Rating or Review Input

Trigger: At Step 5 of the Basic Flow, server-side input validation detects that the rating score is missing/out of range or the review text violates input validation rules.

1. The system rejects the submission without saving the record to the database.
2. The system displays clear validation messages indicating the specific fields that need correction.
3. The Customer corrects the rating or review text and resubmits the form, resuming at Step 4 of the Basic Flow; otherwise, the use case ends.

14.2.2.3 Duplicate Review Submission for the Same Movie

Trigger: At Step 2 or Step 5 of the Basic Flow, the backend detects that the eligible Customer has already submitted a review/rating for the selected movie.

1. The system prevents creating a duplicate review entry for the same Customer and movie.
2. The system either allows the Customer to update their existing review/rating (re-validating at Step 5) or informs the Customer that their feedback has already been recorded.

3. The use case ends.

14.2.2.4 Unauthenticated User Attempts to Submit a Review

Trigger: At Step 1 of the Basic Flow, an unauthenticated visitor attempts to submit a movie review or rating.

1. The system blocks the unauthenticated submission and prompts the user to log in (Register, Log In, and Manage Profile).
2. If the Customer logs in successfully, the flow resumes at Step 2 of the Basic Flow to check eligibility; otherwise, the use case ends.

14.3 Special Requirements

14.3.1 Server-Side Eligibility Enforcement and Input Validation

The backend must enforce role-based access control and verify Customer eligibility server-side over HTTPS before accepting any movie review or rating. All user-supplied review text and rating values must be validated on the backend.

14.3.2 Responsive Usability and Clear Feedback

The review and rating interface must be responsive across mobile phones, tablets, and desktop browsers (Chrome, Edge, Firefox), providing readable contrast, clear labels, and understandable validation and confirmation messages.

14.3.3 Data Privacy

Displayed movie reviews must expose only necessary public display information and must never expose sensitive Customer account credentials, payment details, or private booking identifiers.

14.4 Preconditions

14.4.1 Active Movie Record

The target movie exists in the MEDI-CINE catalogue.

14.4.2 Authenticated and Eligible Customer

The Customer is logged into an active Customer account and meets the system's booking/attendance eligibility criteria for the movie.

14.5 Postconditions

14.5.1 Success End Condition

The Customer's movie rating and review are stored in the authoritative backend database and are visible on the movie details page.

14.5.2 Failure / Ineligible End Condition

If the Customer is unauthenticated, ineligible, or submits invalid inputs, no review record is created and existing movie rating data remains unchanged.

14.6 Extension Points

14.6.1 Browse Movies and View Movie Details

Location: Basic Flow, Step 7 (published ratings and reviews are displayed to Customers browsing the movie details page).

15. Manage Promotions and Discount Codes

15.1 Manage Promotions and Discount Codes

15.1.1 Brief Description

Primary Actor: Administrator
The Administrator uses the desktop administration dashboard to create, view, update, activate, or deactivate promotional rules and discount codes. The system enforces role-based access control, validates promotion parameters server-side, and stores the authoritative promotional rules and discount codes in the backend database so they can be validated and applied during customer ticket booking.

15.2 Flow of Events

15.2.1 Basic Flow

1. This use case begins when an authenticated Administrator navigates to the Promotions and Discount Codes Management section on the desktop administration dashboard.
2. The system verifies the Administrator's role permissions over HTTPS and retrieves the current list of promotional rules and discount codes from the backend database, displaying each promotion's code/name, discount value/rule, validity period, usage conditions, and active status.
3. **Create a New Promotion or Discount Code:**

- a. The Administrator selects the option to create a new promotion or discount code.
- b. The system displays the promotion configuration form.
- c. The Administrator enters the promotional details (including the discount code identifier, promotional rules, discount value, validity start/end dates, and activation status) and submits the form.
- d. The system validates the input fields on the backend, confirms that the discount code identifier is unique, saves the new promotion record in the authoritative database, logs the change, and displays a confirmation message.

4. **Update an Existing Promotion or Discount Code:**

- a. The Administrator selects an existing promotion or discount code from the list and chooses the edit option.
- b. The system displays the promotion's current rules and parameters in an editable form.
- c. The Administrator modifies the promotional rules, validity dates, or status and submits the changes.
- d. The system validates the updated parameters on the backend, updates the promotion record in the database for subsequent booking validations, logs the state change, and displays a confirmation message.

5. **Deactivate a Promotion or Discount Code:**

- a. The Administrator selects an active promotion or discount code and chooses to deactivate it.
- b. The system updates the promotion's status to inactive in the backend database so it can no longer be applied to new bookings, while preserving historical booking and payment records that previously used the code.

15.2.2 Alternative Flows

15.2.2.1 Invalid Promotion Parameters or Missing Required Fields

Trigger: At Step 3d or Step 4d of the Basic Flow, server-side input validation detects missing required fields or invalid promotional rules (e.g., an empty discount code, a negative or excessive discount value, or an end date earlier than the start date).

1. The system rejects the submission without modifying the database.
2. The system displays clear validation messages indicating the specific fields or rules that must be corrected.
3. The Administrator corrects the inputs and resubmits the form, resuming at Step 3c or Step 4c; otherwise, the use case ends.

15.2.2.2 Duplicate Discount Code Identifier

Trigger: At Step 3d or Step 4d of the Basic Flow, the backend detects that the submitted discount code string already exists for another promotion record.

1. The system prevents the creation of a duplicate discount code and displays a clear validation message.
2. The Administrator enters a unique discount code identifier or edits the existing promotion record instead, resuming at Step 3c or Step 4.

15.2.2.3 Deactivating or Modifying a Promotion While Bookings Are Active

Trigger: At Step 4d or Step 5b of the Basic Flow, the Administrator updates or deactivates a discount code while there are existing PENDING or SUCCESS bookings that have already applied the code.

1. The system applies the updated rules or deactivation strictly to subsequent discount-code validation attempts on new bookings.
2. The system preserves the already committed price and discount amounts on existing PENDING or SUCCESS bookings, issued QR e-tickets, and revenue reports to maintain operational and financial consistency.
3. The system completes the update and displays a confirmation message to the Administrator.

15.2.2.4 Unauthorized Role Access Attempt

Trigger: At Step 1 of the Basic Flow (or via direct backend API call), an unauthenticated user, a Customer, or a Cinema Staff member attempts to create, edit, or deactivate promotional rules and discount codes.

1. The backend role-based access control mechanism denies the request.
2. The system displays an access-denied message and logs the unauthorized attempt without altering any promotion records.
3. The use case ends.

15.3 Special Requirements

15.3.1 Authoritative Server-Side Validation and Pricing Consistency

The backend system must remain the single authoritative source for promotional rules, discount-code validation, and final booking price calculations. Modifying or deactivating a promotion must never corrupt historical booking totals or revenue reports.

15.3.2 Role-Based Access Control and Input Validation

All administrative endpoints for managing promotions and discount codes must be restricted strictly to authenticated Administrator roles over HTTPS, with server-side validation for all inputs.

15.3.3 Desktop Dashboard Compatibility and Usability

The promotions management interface must be optimized for desktop browsers supporting current versions of Chrome, Edge, and Firefox, providing readable contrast, clear labels, and understandable validation messages.

15.3.4 Observability and Auditability

All creations, modifications, and deactivations of promotional rules and discount codes must be logged by the backend without logging sensitive secrets.

15.4 Preconditions

15.4.1 Authenticated Administrator Session

The Administrator is logged into the MEDI-CINE desktop administration dashboard with an active, role-verified Administrator session.

15.4.2 Operational Backend and Database

The backend REST APIs and database persistence layer are operational over HTTPS.

15.5 Postconditions

15.5.1 Success End Condition

The created, updated, or deactivated promotional rule and discount code are persisted in the authoritative backend database and are immediately enforced when Customers validate discount codes during ticket booking (Book Movie Tickets).

15.5.2 Failure / Rejection End Condition

If input validation fails, a duplicate discount code is detected, or access is unauthorized, no invalid changes are saved and existing promotional rules remain unchanged.

15.6 Extension Points

15.6.1 Book Movie Tickets (Discount Code Validation)

Location: Basic Flow, Steps 3d and 4d (active discount codes and promotional rules configured in this use case are evaluated when a Customer enters a discount code during booking creation).

15.6.2 Send Notifications and Reminders

Location: Basic Flow, Step 3d (when a new promotion is activated, the system may notify Customers of the new promotion).

16. View Notifications and Reminders

16.1 View Notifications and Reminders

16.1.1 Brief Description

Primary Actor: Customer
Secondary Actor: Notification / Email Service
The system generates and delivers notifications and reminders to the Customer-covering booking confirmations, payment updates, e-ticket notifications, upcoming-showtime reminders, and new promotions-via the customer web interface and connected notification/email service. An authenticated Customer views their notifications and reminders to stay informed of booking statuses, access issued QR e-tickets, and prepare for upcoming screenings.

16.2 Flow of Events

16.2.1 Basic Flow

1. This use case begins when a notification-triggering event occurs in the system (such as a booking confirmation, a payment state update, QR e-ticket issuance, a scheduled reminder window before an upcoming showtime, or a new promotion announcement), or when an authenticated Customer opens the Notifications section on the customer web interface.
2. The backend generates a notification record derived from the authoritative booking, payment, e-ticket, showtime, or promotion data, stores it for the Customer's account, and dispatches the message through the configured notification/email service.
3. The authenticated Customer accesses the Notifications view on the responsive customer web application (or opens the delivered email/notification).
4. The system verifies the Customer's session over HTTPS and retrieves the chronological list of notifications and reminders associated with the Customer's account.
5. The system displays the notification list with clear labels and status indicators, distinguishing between booking confirmations, payment updates, e-ticket notifications, upcoming-showtime reminders, and promotional announcements.
6. The Customer selects a specific notification or upcoming-showtime reminder to view its full details.
7. The system marks the selected notification as read and displays the complete message details, including movie title, screening room, showtime, seat numbers, payment status, or promotional rules as applicable.
8. If the selected item is an e-ticket notification, booking confirmation, or upcoming-showtime reminder for a confirmed (SUCCESS) booking, the Customer may select the direct link to view the QR e-ticket or booking details.

16.2.2 Alternative Flows

16.2.2.1 No Notifications or Reminders Available

Trigger: At Step 4 of the Basic Flow, the backend database query finds no notifications or reminders associated with the Customer's account.

1. The system displays a clear message indicating that the Customer currently has no notifications or reminders.
2. The Customer may continue browsing movies (Browse Movies and View Movie Details), and the use case ends.

16.2.2.2 External Notification / Email Delivery Failure

Trigger: At Step 2 of the Basic Flow, the external notification or email service is temporarily unreachable or fails to deliver an outbound message.

1. The backend logs the delivery error without exposing sensitive user data or system secrets, and without

interrupting or rolling back the core booking, payment, or e-ticket issuance transaction.
2. The system retains the generated notification record in the backend database so the Customer can still view the booking confirmation, payment update, e-ticket, or reminder directly inside the MEDI-CINE web interface and booking history.
3. The flow continues at Step 3 of the Basic Flow when the Customer accesses the web interface.

16.2.2.3 Upcoming Showtime Is Cancelled or Ticket Already Used Prior to Reminder

Trigger: At Step 1 or Step 2 of the Basic Flow, the scheduled reminder timer triggers before a showtime, but the backend check determines that the booking is no longer active/unused (e.g., the showtime was cancelled or the ticket is already marked USED).

1. If the showtime was cancelled, the system suppresses the standard upcoming-showtime reminder and instead ensures a cancellation/status update notification is recorded for the Customer.
2. If the booking was never paid (FAILED or EXPIRED), no upcoming-showtime reminder is sent.
3. The use case ends.

16.2.2.4 Unauthenticated User Attempts to View Account Notifications

Trigger: At Step 3 of the Basic Flow, an unauthenticated user attempts to open the personal Notifications view.

1. The system denies unauthenticated access and redirects the user to the login page (Register, Log In, and Manage Profile).
2. Once the Customer logs in, the flow resumes at Step 4 of the Basic Flow; otherwise, the use case ends.

16.3 Special Requirements

16.3.1 Consistency with Authoritative Booking, Payment, and Ticket States

All notifications and reminders must reflect the authoritative backend state. E-ticket notifications and upcoming-showtime reminders must be generated strictly for confirmed (SUCCESS) bookings with valid tickets, and must never issue or link to an unpaid QR ticket.

16.3.2 Fault Isolation from Core Booking Workflow

Notification or email dispatch failures must be handled gracefully so that they never block seat holds, payment confirmation, or QR e-ticket retrieval in the core booking workflow.

16.3.3 Security, Privacy, and Role-Based Access Control

Personal booking, payment, and e-ticket notifications must be accessible only to the authenticated owning Customer over HTTPS. The system must collect and transmit only information needed for accounts and bookings without logging secrets.

16.3.4 Responsive Usability Across Devices

Notification lists and reminder messages must render with readable contrast and clear labels across mobile phones, tablets, and desktop browsers (Chrome, Edge, Firefox).

16.4 Preconditions

16.4.1 Authenticated Customer Account

The Customer has a registered account in the MEDI-CINE system and, for viewing in-app notifications, an active authenticated Customer session.

16.4.2 Operational Backend and Notification Service

The backend REST APIs, database persistence layer, and notification/email service integration are configured.

16.5 Postconditions

16.5.1 Success End Condition

Relevant booking confirmations, payment updates, e-ticket notifications, upcoming-showtime reminders, or

promotion notices are recorded and viewed by the Customer, and the Customer can navigate directly to their confirmed e-ticket or booking details.

16.5.2 Data Integrity (No Corruption of Operational State)

Viewing notifications or experiencing an external email/notification delivery error does not alter authoritative seat availability, booking statuses, payment records, or ticket USED states.

16.6 Extension Points

16.6.1 Retrieve E-Ticket / View Booking History

Location: Basic Flow, Step 8 (when the Customer clicks on a booking confirmation, e-ticket notification, or upcoming-showtime reminder to open the QR e-ticket or booking details).

16.6.2 Browse Movies and View Movie Details

Location: Basic Flow, Step 8 (when the Customer selects a promotion notification to browse eligible movies and showtimes and apply the discount code during booking).

16.6.3 PWA Push / Offline-Adjacent Notifications (Optional Extension)

Location: Basic Flow, Step 2 (if the optional PWA support feature is implemented after the 14 core capabilities are stable, notifications may also be surfaced via browser/PWA notification channels on supported devices).

17. Use AI Movie and Showtime Assistant

17.1 Use AI Movie and Showtime Assistant

17.1.1 Brief Description

Primary Actor: Customer
Secondary Actor: External AI / LLM Service
The Customer submits a natural-language query to the optional AI movie and showtime assistant on the customer web interface to discover movies or find suitable showtimes. The system validates and sanitizes the request, interacts with an external AI/LLM service through controlled APIs without exposing personal user data, grounds the recommendations strictly in authoritative MEDI-CINE catalogue and showtime data, and presents matching movies and showtimes so the Customer can proceed to movie details or seat selection.

17.2 Flow of Events

17.2.1 Basic Flow

1. This use case begins when a Customer opens the AI Movie and Showtime Assistant panel on the customer-facing web interface.
2. The system displays the assistant prompt along with example natural-language queries (such as searching by genre, mood, age suitability, or preferred screening date/time).
3. The Customer enters a natural-language request asking for movie recommendations or available showtimes and submits the query.
4. The backend validates the input, strips any unnecessary personal or session data, and processes the query using the controlled AI/LLM integration grounded in approved MEDI-CINE movie catalogue, room format, and active showtime records.
5. The system verifies the recommended movies and showtimes against the authoritative backend database to ensure that only active movies (now-showing or coming-soon) and valid, currently scheduled showtimes are included.
6. The system displays the AI-generated response along with structured cards/links for the matching movies and available showtimes.
7. The Customer selects a recommended movie to view its full details (transitioning to Browse Movies and View Movie Details) or selects a recommended showtime to proceed directly to seat selection (transitioning to Select Seats).

17.2.2 Alternative Flows

17.2.2.1 No Matching Movies or Showtimes Found in System Data

Trigger: At Step 4 or Step 5 of the Basic Flow, the Customer's natural-language criteria do not match any active movies or scheduled showtimes in the authoritative MEDI-CINE database.

1. The system prevents the assistant from fabricating non-existent movies or schedules.
2. The system displays a polite response explaining that no current movies or showtimes match the exact request, and optionally suggests closely related now-showing movies from the approved catalogue or invites the Customer to use standard catalogue search/filters.
3. The Customer may refine their natural-language query at Step 3 or switch to standard browsing; otherwise, the use case ends.

17.2.2.2 Out-of-Scope, Unsupported, or Invalid Query

Trigger: At Step 4 of the Basic Flow, the Customer submits an empty input, an overly long/malformed prompt, or a request unrelated to MEDI-CINE movies and showtimes (or attempts to perform direct seat locking/payment through chat).

1. The backend input validation or controlled prompt filter rejects the out-of-scope request.
2. The system displays a clear validation message informing the Customer that the assistant only supports movie and showtime discovery within MEDI-CINE.
3. The Customer may enter a valid movie/showtime query at Step 3, or exit the assistant.

17.2.2.3 External AI / LLM Service Unavailable or Timed Out (Graceful Fallback)

Trigger: At Step 1 or Step 4 of the Basic Flow, the external AI/LLM API is unreachable, times out, or is disabled in the environment.

1. The backend logs the AI integration error without exposing API keys/secrets and without affecting the core movie catalogue or booking services.
2. The system notifies the Customer that the AI assistant is temporarily unavailable and directs the Customer to the standard movie search and filter interface (Browse Movies and View Movie Details).
3. The core booking workflow remains completely unaffected, and this use case ends.

17.3 Special Requirements

17.3.1 Grounding in Authoritative Backend Data (No Fabricated Data)

All movie titles, genres, age ratings, showtimes, and prices referenced by the assistant must be grounded strictly in approved MEDI-CINE backend data. The AI service has limited permissions and may never directly modify seat holds, bookings, prices, or payments.

17.3.2 Privacy and Secret Protection

The system must never send unnecessary personal Customer data (such as passwords, payment details, or private account identifiers) to the external AI/LLM service, and AI API keys/secrets must remain protected on the backend over HTTPS.

17.3.3 Modular Isolation and Non-Disruption of Core Workflow

The AI integration must be isolated in a separate module so that any latency, quota limit, or outage in the AI service never degrades the 14 core capabilities (Browse Movies, Select Seats, Book Movie Tickets, Make Payment, and Validate E-Ticket).

17.3.4 Responsive Usability

The assistant interface must be responsive across mobile phones, tablets, and desktop browsers (Chrome, Edge, Firefox) with readable contrast and clear navigation links to the selected movie or showtime.

17.4 Preconditions

17.4.1 Active Catalogue and Showtime Data

Movie records and active showtimes have been configured in the MEDI-CINE backend database.

17.4.2 Enabled AI Assistant Module and API Connectivity

The optional AI assistant feature is enabled in the deployment environment and connected to the approved AI/LLM API over HTTPS.

17.5 Postconditions

17.5.1 Success End Condition

The Customer receives relevant movie or showtime recommendations grounded in real MEDI-CINE data and can navigate directly to the movie details page or seat selection page.

17.5.2 Data Integrity and Privacy End Condition

No operational database records (movies, showtimes, seat states, bookings, or payments) are altered by the AI assistant, and no unnecessary personal Customer data is transmitted to the external AI provider.

17.6 Extension Points

17.6.1 Browse Movies and View Movie Details

Location: Basic Flow, Step 7 (when the Customer clicks a recommended movie from the AI assistant's response to view its full details, trailer, and reviews).

17.6.2 Select Seats

Location: Basic Flow, Step 7 (when the Customer selects a specific recommended showtime from the AI assistant's response to view the real-time seat map and hold seats).

*USE CASE DIAGRAMS

Diagram 0: High-Level System Package Overview

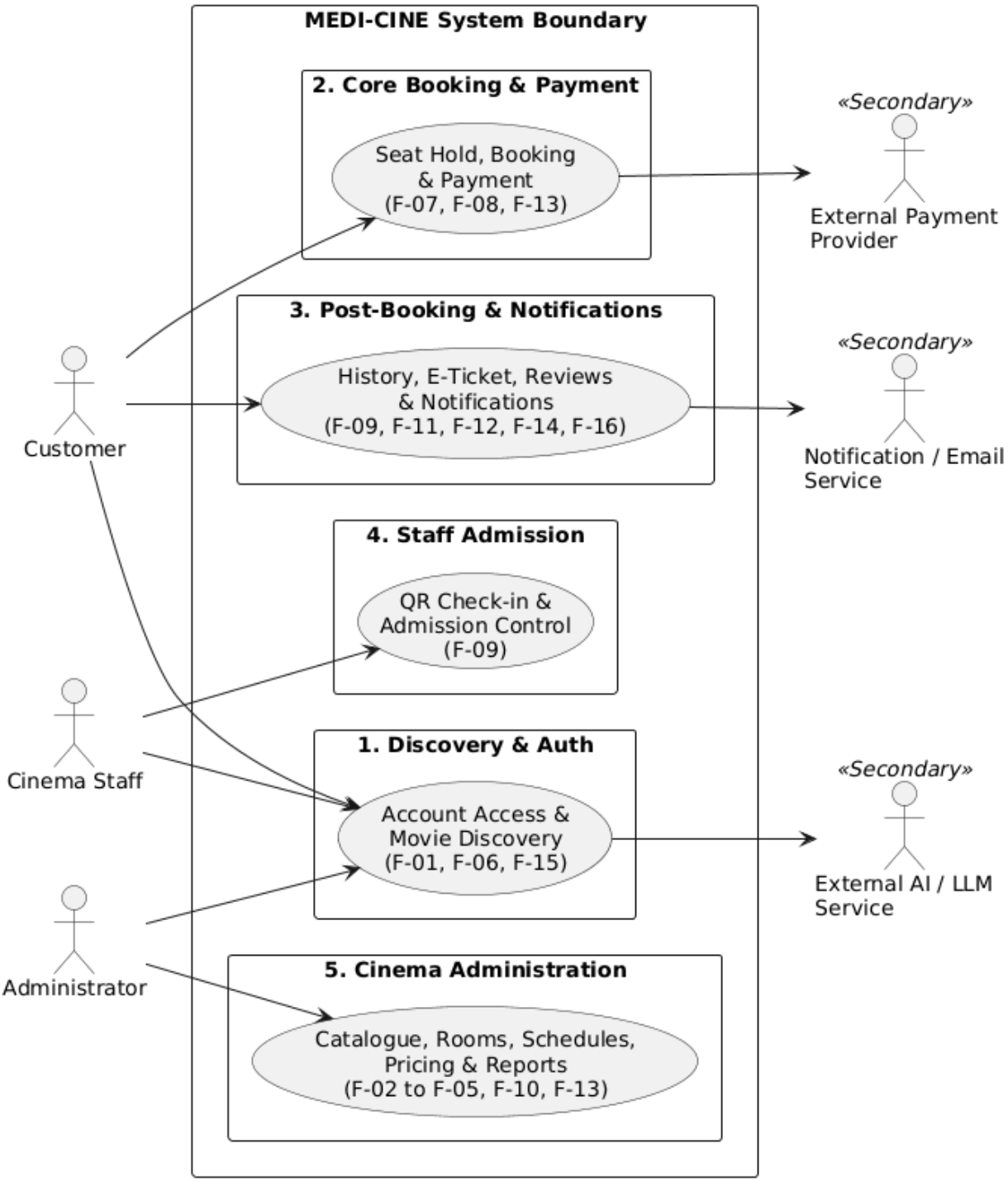


Diagram 1: Authentication & Movie Discovery

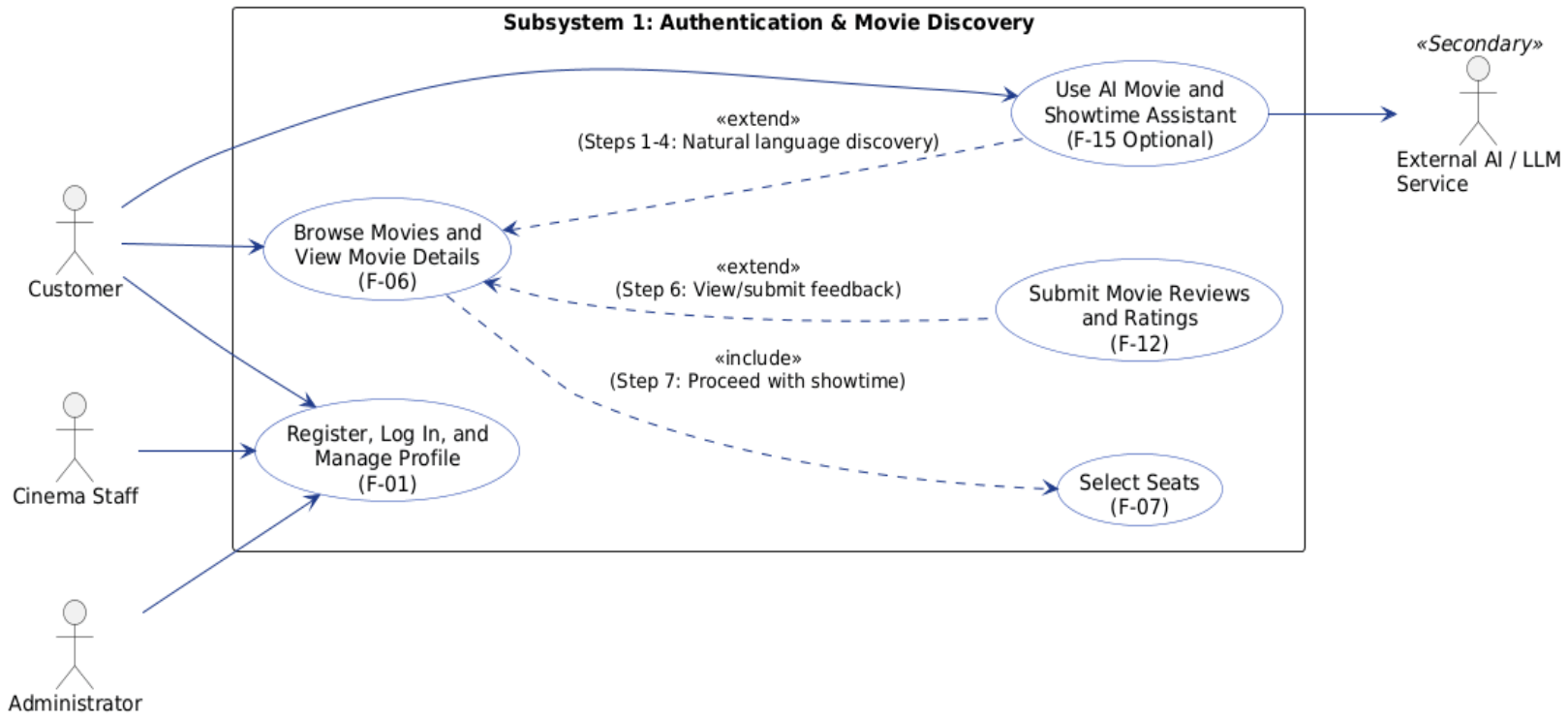


Diagram 2: Core Ticket Booking & Payment Flow

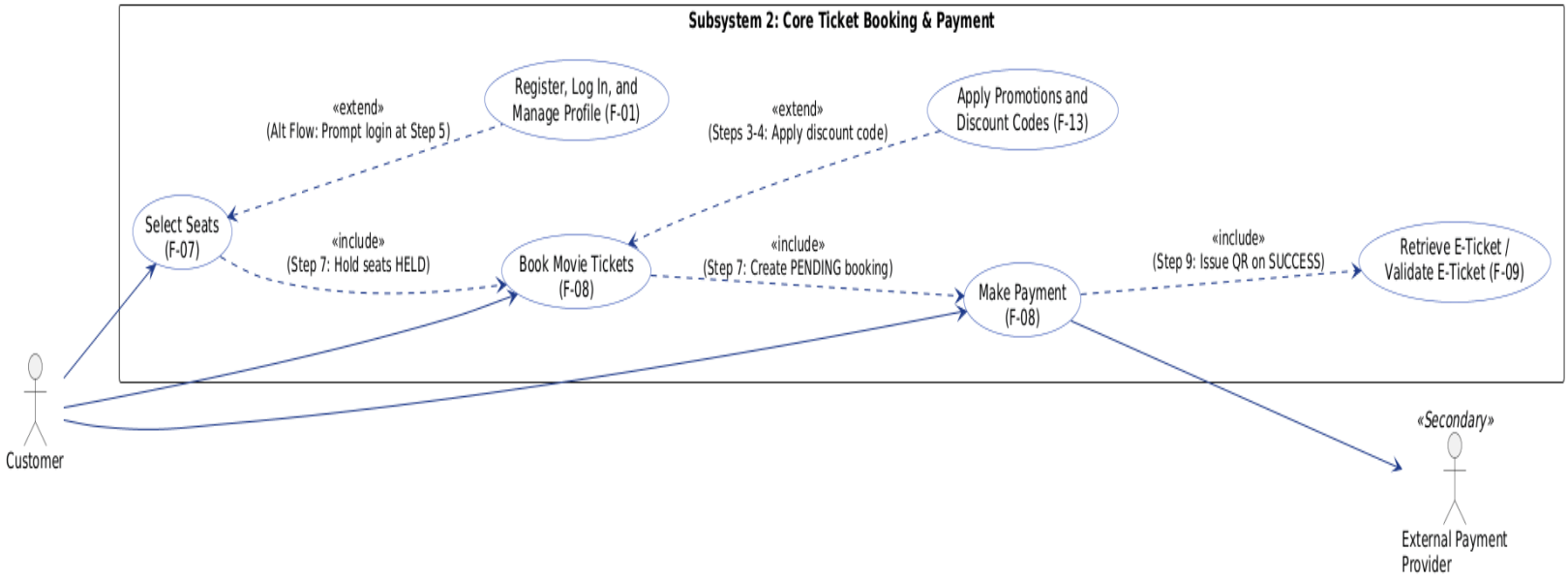


Diagram 3: Customer Post-Booking, History & Notifications

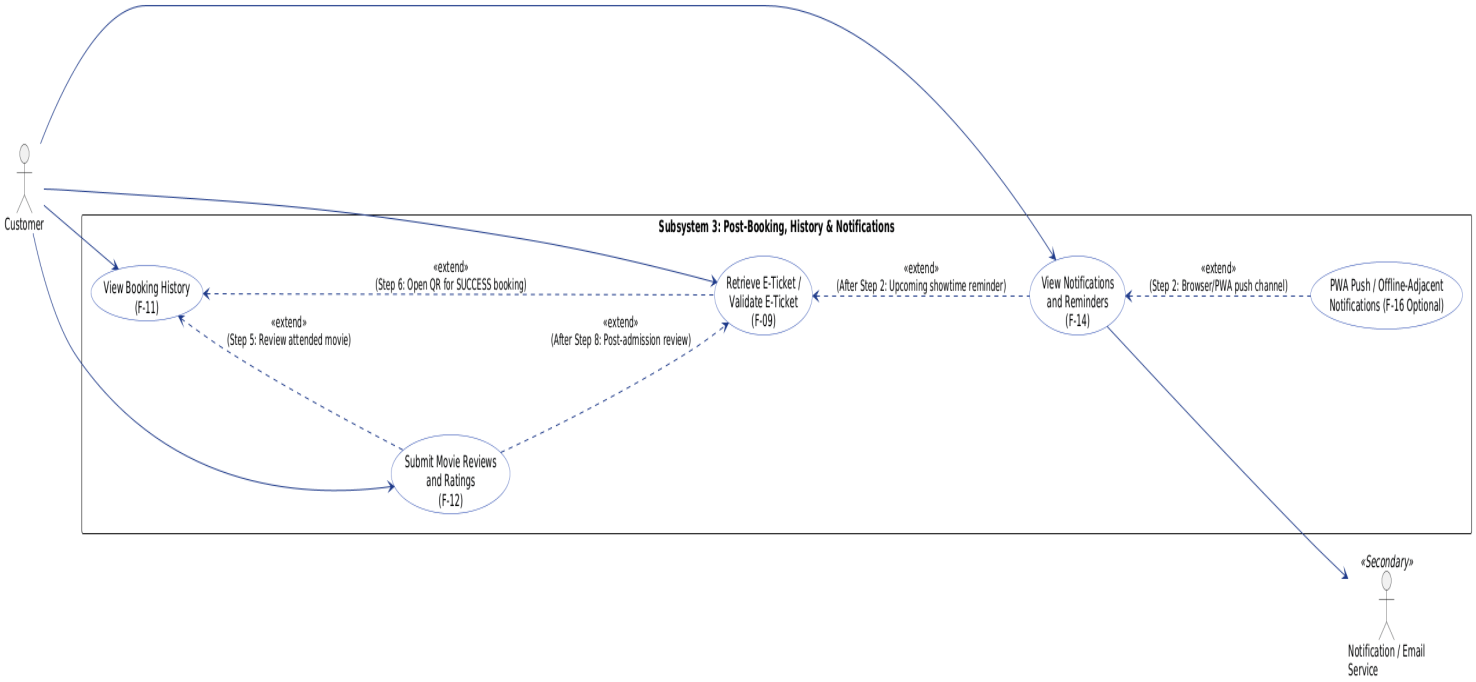


Diagram 4: Cinema Staff E-Ticket Validation & Admission

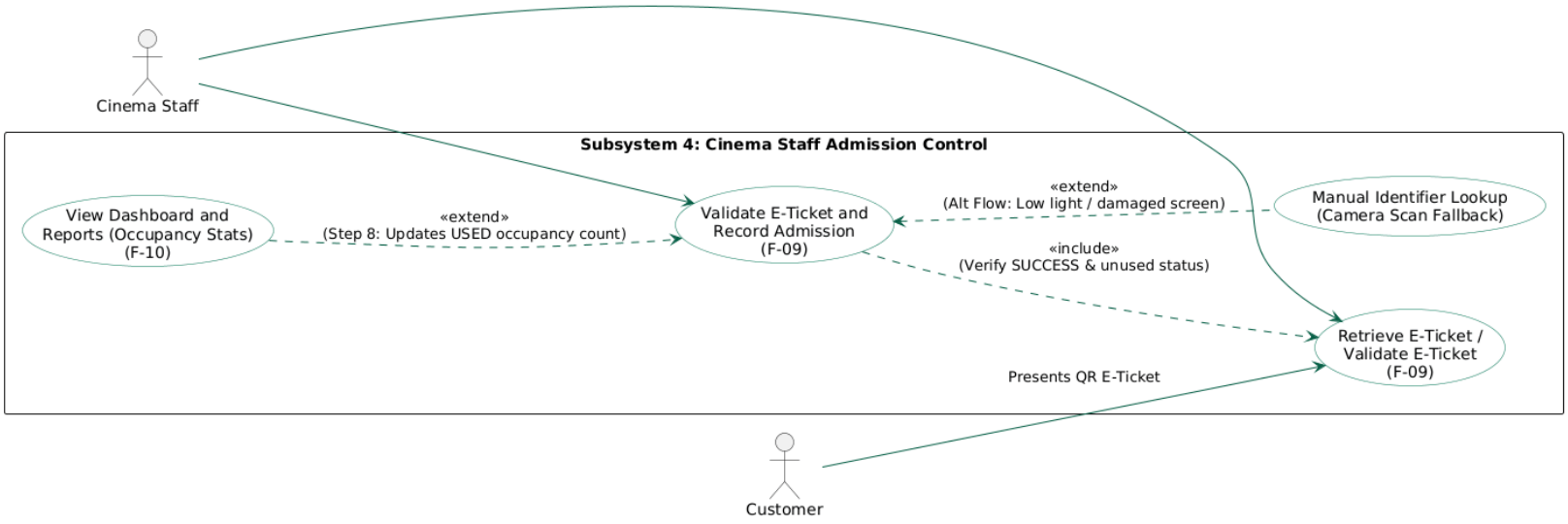


Diagram 5: Cinema Administration & Back-Office Management

